Precambrian Research 324 (2019) 126–145

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

1.86 Ga key paleomagnetic pole from the Murmansk craton intrusions – Eastern Murman Sill Province, NE Fennoscandia: Multidisciplinary approach and paleotectonic applications

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ARTICLE INFO

Keywords:
Fennoscandian Shield
Murmansk craton
Key paleomagnetic pole
Geochronology
Geochemistry
Paleointensity

ABSTRACT

We present the first 1.86 Ga paleomagnetic key pole of Fennoscandia obtained for the dolerite sills of the Murmansk craton – Eastern Murman Sill Province, that outcrop in the northern part of the Kola Peninsula along the Barents Sea coast for a distance of 200 km (Slat = 68.5°; Slong = 37.9°; N = 16 sites; Plat = 54.7°; Plong = 234.7°; dp/dm = 4.3°/6.3°, Qv = 5). The age of the sills and their characteristic remanent magnetization (ChRM) was determined by four independent geochronometers: U-Pb – 1860 ± 4 and 1863 ± 7 Ma (ID-TIMS, baddeleyite), Sm-Nd – 1889 ± 57 Ma, Rb-Sr – 1850 Ma, Ar/Ar – 1865 ± 8 and 1857 ± 20 Ma (biotite). The primary nature of the ChRM is confirmed by the results of petrographic, geochemical, paleo- and rock magnetic studies, as well as by thermochronological data. The similarity of the petrographic and geochemical characteristics of sills from different localities indicates that these dolerite sills were formed during a single magmatic event and their cooling down to 580 °C occurred at depths of about 10 ± 2 km and lasted ~2800 years or even faster. Paleogeographic reconstruction of Fennoscandia on the basis of the obtained paleomagnetic pole is in general agreement with the previously suggested configuration of core of the Nuna/Columbia supercontinent (Evans and Mitchell, 2011; Meert and Santosh, 2017). A new reliable Thellier-Coe paleointensity determination for this time reveals a rather low mean VDM = 1.8 (± 0.1) × 10^{22} Am^2 that supports the Proterozoic dipole low hypothesis (Biggin et al., 2009).

1. Introduction

The validity and reliability of paleocontinental reconstructions depend on using multiple methods, but among these approaches the paleomagnetic method is particularly important. However, the potential of paleomagnetism for the Early Precambrian rocks is often dramatically restricted, primarily due to the partial or complete loss of the primary magnetic record during the rocks' life, as well as because of difficulties with dating the characteristic magnetization.

As it was shown, lithostratigraphic, tectonothermal, geochronological and also paleomagnetic data give information relevant to justify a consistent history and configuration of the Late Precambrian

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https://doi.org/10.1016/j.precamres.2019.01.017
Received 30 October 2018; Received in revised form 27 December 2018; Accepted 29 January 2019
Available online 30 January 2019
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supercontinent Rodinia (Li et al., 2008). The tectonic history of the Early Precambrian supercontinent Nuna (Columbia) is controversial mostly due to significantly less reliable paleomagnetic data (Kaur et al., 2013; Nance et al., 2014; Piper, 2013; Roberts, 2013; Yakubchuk, 2010; Zhang et al., 2012). Available geological and geochronological data suggest that Nuna amalgamation was related with numerous ca. 2.1–1.8 Ga collisional orogenic belts, that can be found in most early Precambrian cratons (Zhao et al., 2002). The peak of Nuna packing occurred at 1.90–1.85 Ga based on ages of collisional orogens (Rogers and Santosh, 2009). The most extensive orogenesis of this age is manifested in the North American, North Atlantic and Siberian cratons, and Fennoscandia, considered as the main components of the Nuna's core (Evans and Mitchell, 2011; Meert and Santosh, 2017). These blocks have a close relative position in different models of the Nuna supercontinent (Kaur et al., 2013; Lahtinen et al., 2009; Piper, 2013; Rogers and Santosh, 2002; Yakubchuk, 2010; Zhang et al., 2012) and assembly into Nuna is based on correlation of lithostratigraphic and geochronological data from 1.90 to 1.85 Ga collisional orogens and related cratons (Lahtinen et al., 2009; Lebrun et al., 2018; Zhao et al., 2002). However, the paleomagnetic data for this time interval are extremely scarce and heterogeneous (Mitchell et al., 2010).

The main formation stages of the Earth as a planet, including the emergence of its magnetic field, occurred in the Precambrian. The main characteristics of the geomagnetic field such as geometry, intensity and reversals reflect the geodynamical processes in the Earth's interior. Paleomagnetic studies that explore the evolution of these parameters with time provide extremely valuable information on these processes, including the nature and timing of inner core nucleation. The last question is the most crucial for the modern geology, because existing estimates of the inner core formation range from the Archean to Early Cambrian (Aubert et al., 2009; Driscoll, 2016; Gubbins et al., 2004; Olson et al., 2013). We present new reliable paleointensity data obtained for studied 1.86 Ga dolerite sills and use them to test the Proterozoic dipole low hypothesis (Biggin et al., 2009).

2. Geological setting

The Fennoscandian Shield is the best-exposed and well-studied crustal segment of the East European craton (Bogdanova, 1993) and comprises two different parts. South-western Fennoscandia consists of large Paleo- and Mesoproterozoic provinces of juvenile igneous rocks (Fig. 1a), formed during a number of successive orogenic events of age 1.93–0.92 Ga (Bogdanova et al., 2015; Korja et al., 2006; Lahtinen et al., 2005, 2009).

The north-eastern part of the shield is composed of Archean crust, penetrated by Paleoproterozoic mafic intrusions and experienced tectonic and metamorphic reworking related to the Lapland-Kola collisional orogeny (LKO) 1.93–1.87 Ga (Fig. 1a) (Balagansky et al., 1998; Daly et al., 2001, 2006). The Archean crust of the Belomorian and Kola provinces is strongly reworked during this event. The Belomorian province was reworked more intensively (up to eclogite facies (Yu et al., 2017)) and longer (up to 1750 Ma according to titanite U-Pb dating (Bibikova et al., 2001)) than Kola province, and is usually considered as a footwall of the orogenic core (Daly et al., 2006). In the Kola province, considered as a hanging wall of the orogenic core, Archean and Early Paleoproterozoic rocks mostly preserve their primary characteristics (Daly et al., 2006). The Karelian and Murmansk provinces represent inner parts of the Archean crustal blocks located far from the Lapland-Kola orogen core. They are largely free of Paleoproterozoic compressional deformation, and preserve primary characteristics of Archean crustal complexes and are considered as Archean cratonic blocks (Hölttä et al., 2008).

Paleoproterozoic mafic volcanics, dyke swarms and layered intrusions ca. 2.51–1.72 Ga are wide spread in the Eastern Fennoscandia (Hanski, 2013 and references therein). Within-plate mafic igneous rocks of age 2.51–2.06 Ga predate the Lapland-Kola orogeny and are considered as mantle plume-related events. Some of them could be related to a break-up of the Archean blocks and opening of the Lapland-Kola Ocean (Lahtinen et al., 2008; Puchtel et al., 1999). Later events (1.98–1.77 Ga) took place both in Karelian and Kola-Murmansk provinces synchronously with several stages of Lapland-Kola orogeny (Daly et al., 2006).

The Murmansk craton is a narrow (60–70 km in width) segment of the Archean crust traced along the Barents Sea coast for 600 km from Sredniy Peninsula to the east (Fig. 1). It is separated from the adjacent Kola province by a series of deep-seated faults (Glebovitsky, 2005). The Murmansk craton consists of tonalite-trondhjemite-granodiorite (TTG) gneisses with sparse inclusions of mafic rocks, that are considered as the remnants of reworked greenstone belts (Kozlov et al., 2006). Intrusive enderbites, diorites, tonalites, trondhjemites and granites comprise a significant part of the Murmansk craton. The geochronological data for Teriberka (enderbites, 2772 ± 7 Ma) and Iokanga (amphibolites, gneisses, enderbites, diorites, trondhjemites, and granites 2717–2790 Ma) areas suggest that main crustal-forming events in the Murmansk craton occured at 2.7–2.8 Ga (Kozlov et al., 2006). These rocks have $T_{Nd}DM$ values vary from 2.73 to 3.03 Ga (Kozlov et al., 2006; Timmerman and Daly, 1995) suggesting a relatively short crustal residence time for the source of granitoids.

At least five episodes of mafic magmatism of age 2.68, 2.50, 1.98, 1.86 and 0.38 Ga are distinguished in the Murmansk craton according to U-Pb baddeleyite and zircon dating results (Arzamastsev et al., 2009; Fedotov et al., 2012; Stepanova et al., 2018). Studied dolerite sheet intrusions (hereinafter we will call them "sills") in the Murmansk craton (Fig. 1; Supplementary, Fig. S1) have been previously considered as Riphean (Koistinen et al., 2001), but preliminary results of zircon and baddeleyite U-Pb dating prove their Paleoproterozoic ca. 1860 Ma age (Arzamastsev et al., 2009; Fedotov et al., 2012).

3. Sampling and methods

Each sill-like intrusion was studied using a set of methods, including petrological, geochemical and paleomagnetic investigations. Two most representative sills were dated by several isotope methods.

3.1. Petrographic, geochemical and isotopic methods

Petrographic studies were carried out with optical polarizing microscope and Vega II LSH scanning electron microscope equipped with EDS Inca Energy-350 at the Centre for Collective Usage of Scientific Equipment, Karelian Research Center of the Russian Academy of Sciences (CCU KarRC RAS), Petrozavodsk. The composition of minerals was analyzed in polished thin sections with a beam current of 15 nA ± 0.05 nA using a 20 kV accelerating voltage, a working distance of 15 mm, and a counting time of 70 s. X-ray spectra were processed by optimization over spectra using a set of standards of rock-forming minerals. Measurement errors were < 2 rel % for elements with concentrations > 10 wt%; < 5 rel % for elements with concentrations 5–10 wt%, and up to 10 rel % for elements with concentrations 1–5 wt%.

Concentrations of major elements were determined on fused glass discs using a PW-2400 X-ray sequential fluorescence spectrometer at Institute of Geology of Ore deposits, Geochemistry, Mineralogy and Petrography, Russian Academy of Sciences (IGEM RAS), Moscow. The uncertainties were 1–5% for elements with concentrations of > 0.5 wt % and up to 12% for elements with concentrations of < 0.5 wt%. Loss on ignition (LOI) values were determined gravimetrically. Concentrations of trace and rare-earth elements were determined by ICP-MS on a Thermo Scientific X Series 2 instrument at the CCU KRC RAS, using the conventional technique (Svetov et al., 2015). Samples were digested in an acid mixture in open system. The accuracy of the analyses was monitored by the measurement of US Geological Survey reference material BHVO-2 and in-house standards SGD-2a, ST-1a, C-

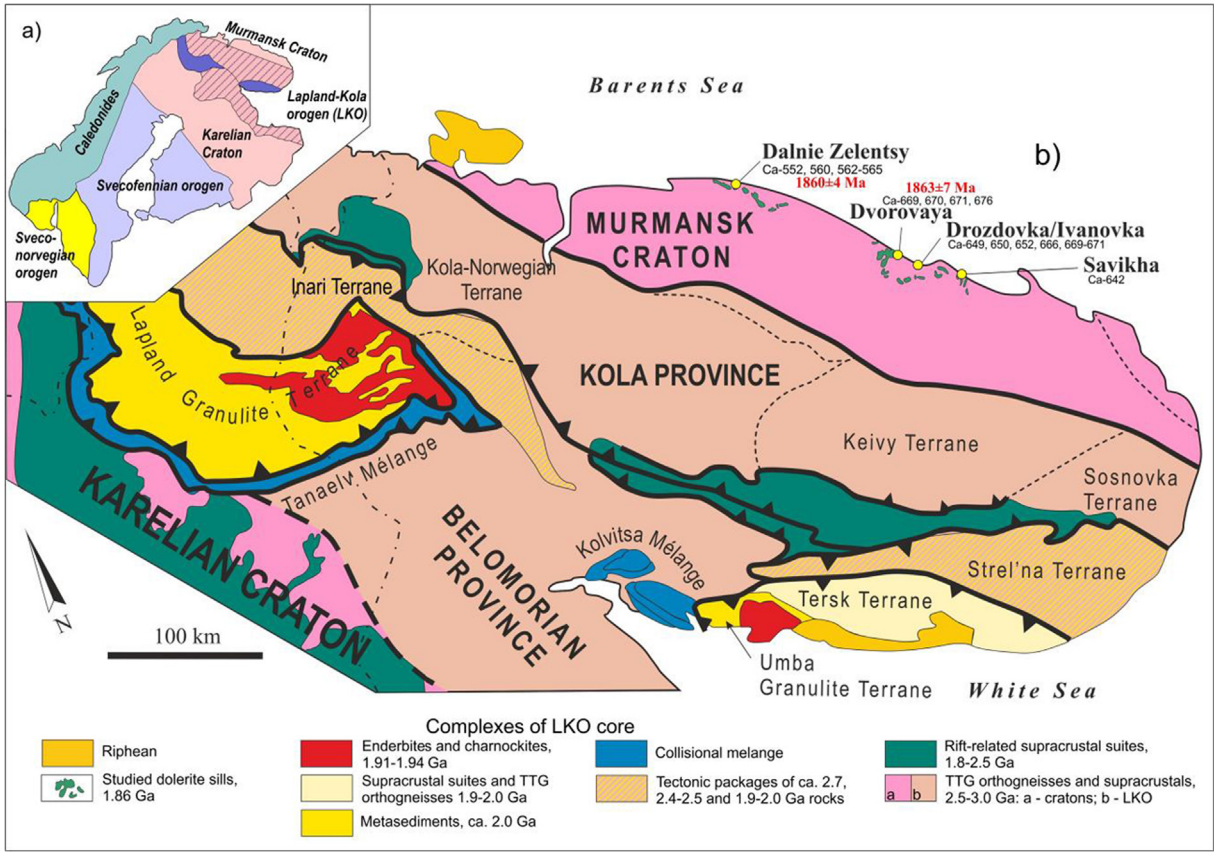


Fig. 1. Sketch tectonic map of the northeastern Fennoscandian Shield and studied localities (yellow circles). Simplified after (Daly et al., 2006). Numbers of the samples are given according to Tables. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

1412. RSD values were less than 5% for most elements, 5–6.5% for Sc, Ti, V, Cr, Co, Ni, Rb, and Cs, and 8.1% for Ti.

The separation of minerals for isotopic studies was carried out at IGEM RAS. Baddeleyite from gabbro-pegmatite samples was extracted using a Wilfley shaking table, following the method described by Söderlund and Johansson (2002). Rock-forming minerals were separated following the standard procedure using density and magnetic separation techniques.

The U-Pb baddeleyite study was carried out at the Laboratory of Isotope Geology, Institute of Precambrian Geology and Geochronology (IPGG RAS) employing the isotope dilution thermal ionization mass-spectrometry technique (ID-TIMS). Hand-picked most transparent and homogeneous baddeleyite grains were ultrasonically washed in ethanol and acetone. Additional soft acid washing to eliminate surface contamination was performed using warm 1 M HNO3 and 1:1 HCl for 20 min each. Then grains (3–10 per analysis) were transferred to a Teflon digestion vessel in the high-pressure, stainless steel-jacketed Teflon bomb (Parr bombs) and spiked with a ^{202}Pb-^{235}U isotopic solution. The grains were dissolved in 10 µl HF at 220 °C overnight. No column chemistry was performed. Isotopic analyses were carried out on a Triton TI multicollector mass spectrometer in digital ion counting mode.

The uncertainties on the U and Pb concentrations were 0.5%, and the blanks were 1–2 pg for Pb and 1 pg for U. The data were reduced using the PbDAT (Ludwig, 1991) and ISOPLOT (Ludwig, 2012) software. Common Pb correction was applied according to the model of (Stacey and Kramers, 1975). All errors are reported as 2σ mean.

The $^{40}Ar/^{39}Ar$ isotope geochronological data were obtained at the Center for collective use of scientific equipment for multi-element and isotope research of the Siberian Branch of the Russian Academy of Sciences (Novosibirsk). Mineral grains for dating were handpicked under a binocular microscope from the 0.25–0.15 mm fraction of crushed rock. The samples were irradiated at the cadmium-lined channel of the VVR-K-type research reactor of the Institute of Physics and Technology at the Tomsk Polytechnic University. Stepwise heating experiments were conducted in an externally heated quartz vessel. The ^{40}Ar blank (20 min at 1200 °C) was no higher than $n \times 10^{-10}$ ncm^3. The extracted argon was purified using Zr-Al SAES getters. Argon isotope composition was measured with a Micromass noble gas 5400 mass spectrometer (Great Britain). Calculated $^{40}Ar/^{39}Ar$ ages are reported with ± 1σ uncertainties.

The Sm-Nd and Rb-Sr isotopic studies were carried out at the Laboratory of Isotope Geochemistry and Geochronology IGEM RAS, following the technique of (Larionova et al., 2007). Measurements of Sr, Nd isotopic composition were performed on a Sector 54 mass-spectrometer (Micromass, the UK), using a multi-dynamic mode routine. The effects of mass-fractionation were corrected using an exponential law, via normalizing to $^{86}Sr/^{88}Sr = 0.1194$, $^{146}Nd/^{144}Nd = 0.7219$. The $^{87}Sr/^{86}Sr$ ratio of the SRM-987 Sr standard measured during the period of the analytical session averaged 0.710242 ± 15 (2SD, N = 31). The mean of the Nd-IGEM in-house Nd standard runs during the period of data collection was 0.512400 ± 10 for $^{143}Nd/^{144}Nd$ (2SD, N = 13, which corresponds to LaJolla Nd standard $^{143}Nd/^{144}Nd = 0.511850$ ± 10). The uncertainty on the $^{147}Sm/^{144}Nd$ ratio was determined by analysis of multiple aliquots of the BCR-1 to be = 0.3% relative (2SD, N = 11). The uncertainty on the $^{87}Rb/^{86}Sr$ is at 1%. $^{147}Sm/^{144}Nd = 0.1967$ $^{143}Nd/^{144}Nd = 0.512638$ were used to calculate the εNd (Jacobsen and Wasserburg, 1984).

3.2. Paleomagnetic, rock magnetic and paleointensity methods

Samples for paleomagnetic and attendant studies were taken as

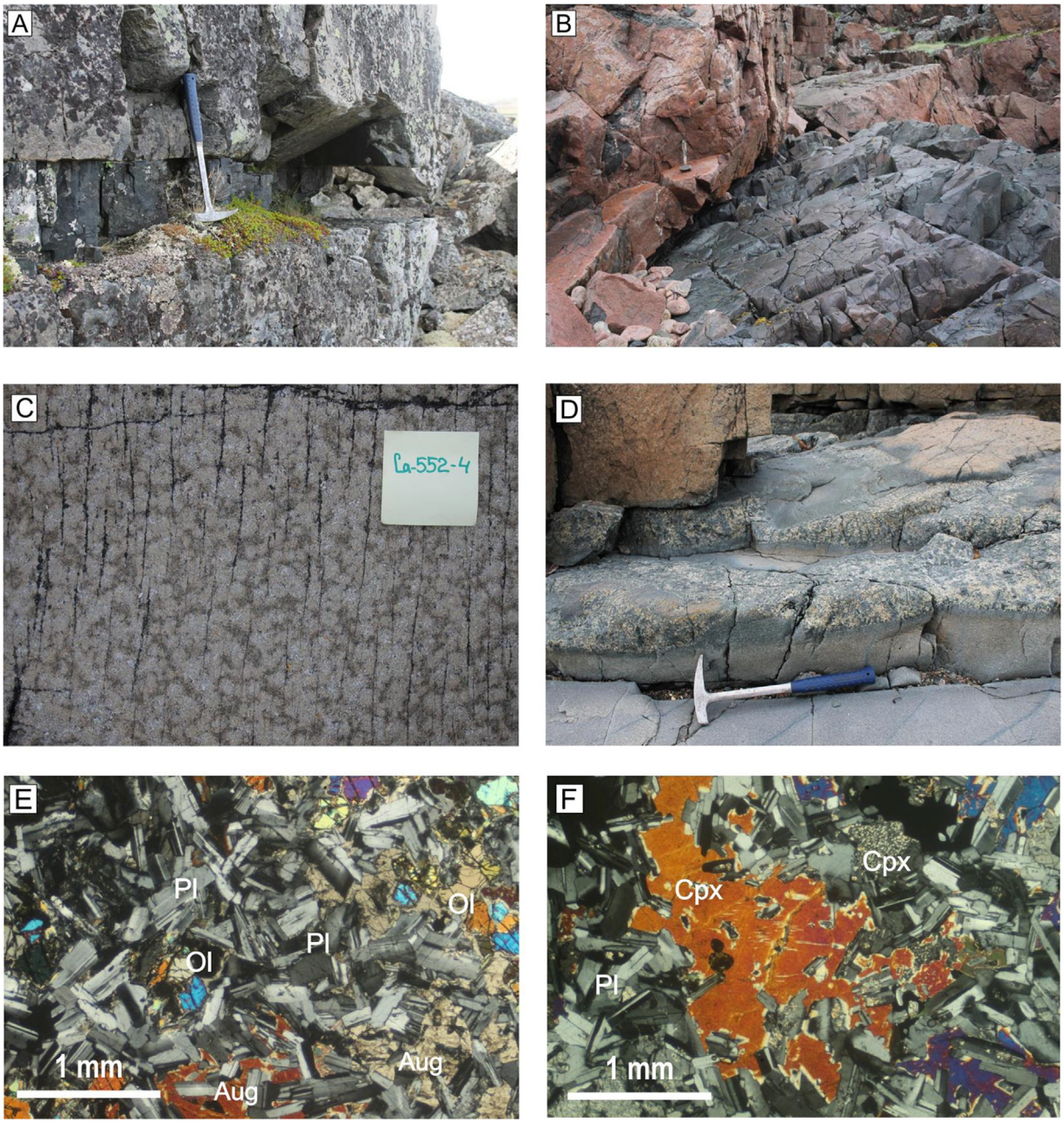


Fig. 2. Photographs illustrate typical features of the ca. 1860 Ma doleritic sills in the Murmansk craton. A – sub-horizontal thin sill in Dalnie Zelentsy locality (Ca-653). Hammer for scale. B – upper contact of inclined sheet intrusion in the Drozdovka locality (Ca-671). Hammer for scale. C – pea-textured dolerite from the central part of the sill, Dalnie Zelentsy locality (Ca-552-4). 8 cm sized paper for scale. D – gabbro-pegmatite vein in Dalnie Zelentsy locality (Ca-552-2). Hammer for scale. E – photomicrograph of olivine-bearing dolerite. Nicols crossed. A – photomicrograph of augite oikocryst, sample Ca-552-4, central part of the sill. Nicols crossed.

block and core samples and oriented using magnetic compass. A sun compass was used to control the influence of disturbing magnetic fields on the magnetic needle when possible. Local declination value was taken according to the IGRF model, the 12th generation. Each thin sill was sampled at one site; thick sills (tens of meters and more) were sampled in two or three sites located at a distance of up to hundreds of meters apart. At least 10 oriented samples were taken from each site, except for the Savikha locality, where only 3 samples were taken.

All specimens were processed at the paleomagnetic laboratories of the Institute of Physics of the Earth Russian Academy of Sciences (IPE RAS, Moscow), Geophysical Observatory (GO) “Borok” IPE RAS, and Lomonosov Moscow State University and heated up to 620 °C in at least 12 steps in MMTD-80A (Magnetic Measurements, UK) thermal demagnetizers. A remanent magnetization of all samples were measured by 2G Enterprises (USA) cryogenic superconducting (SQUID) rock magnetometer (SRM) at the IPE RAS. Sister-samples, represent ~67% of the total collection, were demagnetized by alternating field (AF) up to 130 mT by the build-in demagnetizer of SRM. Some representative samples were subjected to a continuous thermal demagnetization up to 600 °C using the full-vector component vibrating sample magnetometer (3D-VSM) constructed at the GO “Borok”. Directional data were obtained with least squares lines according to standard routines (Chadima and Hrouda, 2006; Enkin, 1994; Kirschvink, 1980). Hysteresis properties, First Order Reversal Curve (FORC) diagrams and temperature dependence of saturation magnetization were made with a vibrating sample magnetometer MicroMag 3900 (Princeton Measurements Corporation, UK) and a variable field translation balance (VFTB, Petersen Instruments, Germany). KLY4 Kappabridge with CS3 furnace apparatus

(AGICO, Chech Rep.) was used to measure the temperature dependence of magnetic susceptibility.

Microscopic investigations were carried out at the GO "Borok" by Olympus-BX51 optical microscope (previewing and selection of etching mode) after which the specimen was studied by Tescan Vega II scanning electron microscope (SEM) with a Drycool block for energy dispersion analysis. After having been previewed in an optical microscope, the specimen was etched by concentrated hydrochloric acid: a 10-s etching was sufficient for revealing the microtopography in the structures of heterophaseoxidized titanomagnetite. After that a conductive carbon coat was thermally sprayed on the specimen. In SEM, the specimen was studied under the accelerating voltage of 20 kV and current ranging from 10 to 200nA.

For some powdered samples, the magnetic fraction, separated with a strong neodymium magnet, was subjected to X-ray structural analysis at the GO "Borok" with a STOE STADI multipurpose diffractometer (cobalt $K\alpha_1$ radiation) at room temperature. From the diffraction spectra, the ferrimagnetic phases were identified with their abundance and lattice constant by comparison with the Crystallography Open Database (Gražulis et al., 2012).

A set of experiments for the investigation of magnetic and thermomagnetic properties of studied rocks and the determination of paleointensity were performed at the GO "Borok". Rock pieces of either drilled cores or hand samples that remained after paleomagnetic experiments were used and several unoriented sister cubes with 1 cm in edge length were cut from them. The thermal stability of magnetic minerals and their Curie points (Tc) were estimated from thermomagnetic curves of strong-field magnetization $M_{si}(T)$ after heating–cooling cycles to incrementally higher temperatures T_i = {200, 300, 400, 500, 600, 700}°C with a Curie balance in an external magnetic field B = 450 mT. Day plot (Day et al., 1977) and additional thermomagnetic criterion (Shcherbakova et al., 2000) were used to assess the domain structure of the ferromagnetic grains.

Paleointensities were determined using two different techniques, described in detail in (Shcherbakova et al., 2017a): (1) the Thellier–Coe procedure (Coe, 1967; Thellier and Thellier, 1959) with the intermediate controlled heatings to lower temperatures – the pTRM-check procedure (Prévot et al., 1985), and (2) the Wilson technique for the rapid estimation of B_{anc} (Wilson, 1961). Two to four sister cube specimens were used from each paleomagnetic sample, the "strongest" of which were subjected to the Thellier-Coe procedure in the 3D-VSM. The other specimens, with magnetic moments insufficient to be measured by 3D-VSM, were heated in the homemade electric furnace with a residual field < 50 nT and measured with the JR-6 magnetometer. No systematic variations of the paleointensity results observed between devices.

4. Results

4.1. Geology and petrography

Numerous dolerite sheet intrusions (sills) outcrop in the northern part of the Murmansk craton along the Barents Sea coast from the Dalnie Zelentsy settlement to Savikha Bay (Fig. 1) covering a total area up to 1000 km^2. Most of intrusions are sub-horizontal, cut Archean granites and gneisses (Fig. 1; Fig. 2, a, b) and vary in thickness from 20 cm up to 100 m. Several sills in the Dalnie Zelentsy, Ivanovka and Drozdovka areas are cut by Devonian tholeiitic and alkali dykes.

Thin sills (less than 20 m in thickness) were studied in the Dalnie Zelentsy, Ivanovka and Drozdovka localities. All these thin sills in these areas are similar in mineral composition and internal structure. Sills with thickness less than 1 m, as well as chilled margins of relatively thick bodies, consist of fine-grained aphyric olivine-bearing dolerites. The main minerals of fine-grained quenched dolerites are plagioclase, augite, and olivine (Fig. 2, e). Biotite, Ti-magnetite, ilmenite, and quartz comprise less than 10 vol% of rocks. Central parts of sills with thickness 1–20 m usually consist of massive homogeneous pea-textured dolerites. Augite forms large (up to 5 mm) skeletal oikocrysts filled by plagioclase chadacrysts. The matrix between large pyroxene grains consists of plagioclase laths, biotite, Ti-Mgt and olivine.

Thick sills (up to 100 m in thickness) also show very restricted variations in mineral composition. They are composed of medium- to coarse-grained dolerites and often contain gabbro-pegmatites in the central parts of intrusions. The pea-textured dolerites (Fig. 2 c, d) are typical for thick sills. Their main minerals are augite (up to 50 vol%) and plagioclase (up to 60 vol%). Olivine, biotite, and Ti-magnetite make up ca. 10 vol% of the rocks. Large laths of Ca-plagioclase often contain small areas with clear margins comprising of sericite, albite, K-feldspar and epidote, that probably represent altered microxenoliths of hosted granite captured at early stages of crystallization. Augite with pigeonite lamellae forms large oikocrysts filled with plagioclase (An_{71-81}). Biotite and Ti-magnetite in the central parts of the sills form relatively large grains containing plagioclase laths.

Gabbro-pegmatites occur as schlieren, lenses, and veins consisting of plagioclase, clinopyroxene, amphibole, biotite, and opaque minerals. Plagioclase and pyroxenes are often replaced by epidote and amphibole, respectively, and opaque minerals often have biotite rims. Quartz, biotite and plagioclase fill interstitials of large grains. Accessory baddeleyite for U-Pb dating was extracted from gabbro-pegmatite schlieren in the Dvorovaya and Dalnie Zelentsy thick sills.

Most sills including Dvorovaya, Savikha, and Dalnie Zelentsy localities are well-preserved. In these rocks plagioclase, pyroxene, and biotite show no alteration (Fig. 2e, f). In the Dalnie Zelentsy locality primary magmatic olivine is associated with thin pyroxene-plagioclase symplectic rims (Supplementary, Fig. S2). These rims probably form at a late-magmatic stages of sill crystallization as in the Skaergaard intrusion (Holness et al., 2011). The fact that symplectic rims are well-preserved indicates the absence of metamorphic alteration. In several cases primary magmatic minerals are altered due to post-magmatic processes: olivine is partly replaced by iddingsite, small laths of plagioclase are partly replaced by sericite, and pyroxenes are surrounded by thin amphibole rim.

Several sills in Ivanovka and Drozdovka localities are intensively altered. The fine-grained chilled dolerites are altered into biotite-amphibole-plagioclase massive rocks, with preserved relicts of doleritic textures. In the central parts of altered sills plagioclase are saussuritized, and pyroxenes replaced by amphibole. Olivine in the central parts of altered sills is partly replaced by fine grains of Fe minerals. One of the thin sills in the Dalnie Zelentsy locality (Ca-564) is also amphibolitized. The widespread distribution of Devonian dykes in Dalnie Zelentsy and Ivanovka-Drozdovka localities suggests that alteration of the studied sills could be related to Devonian magmatism.

4.2. Geochemistry

All of the fine- and medium-grained dolerite sills are very close in chemical composition in all areas. They are moderately magnesian (MgO = 8.0 ± 0.5 wt%), low titanium (TiO_2 = 0.6 ± 0.2 wt%), and high aluminous (Al_2O_3 = 16.0 ± 0.5 wt%) tholeiitic basalts (Fig. 3). The trace element characteristics of these sills are also similar. All samples have moderate concentrations of LILE, HFSE and REE, are slightly enriched in LREE ($[La/Sm]_N$ = 2.6 ± 0.5) and depleted in HREE ($[Gd/Yb]_N$ = 1.4 ± 0.3). Small Eu anomalies (Eu/Eu * = 1.0 ± 0.2) and negative anomalies of all HFSE with most prominent being for Nb (Nb/Nb * = 0.30 ± 0.05) are typical for the studied dolerites (Table 1). Such a homogeneity of geochemical features of the doleritic sills implies their crystallization from a common initial melt.

Two thick (> 30 m) sills from the Dalniye Zelentsy and Dvorovaya localities show wide compositional variations due to the presence of gabbro-pegmatites that have lower MgO content, enriched in LILE, HFSE and REE (Fig. 3; Table 1), and probably represent the residual melts formed via fractional crystallization.

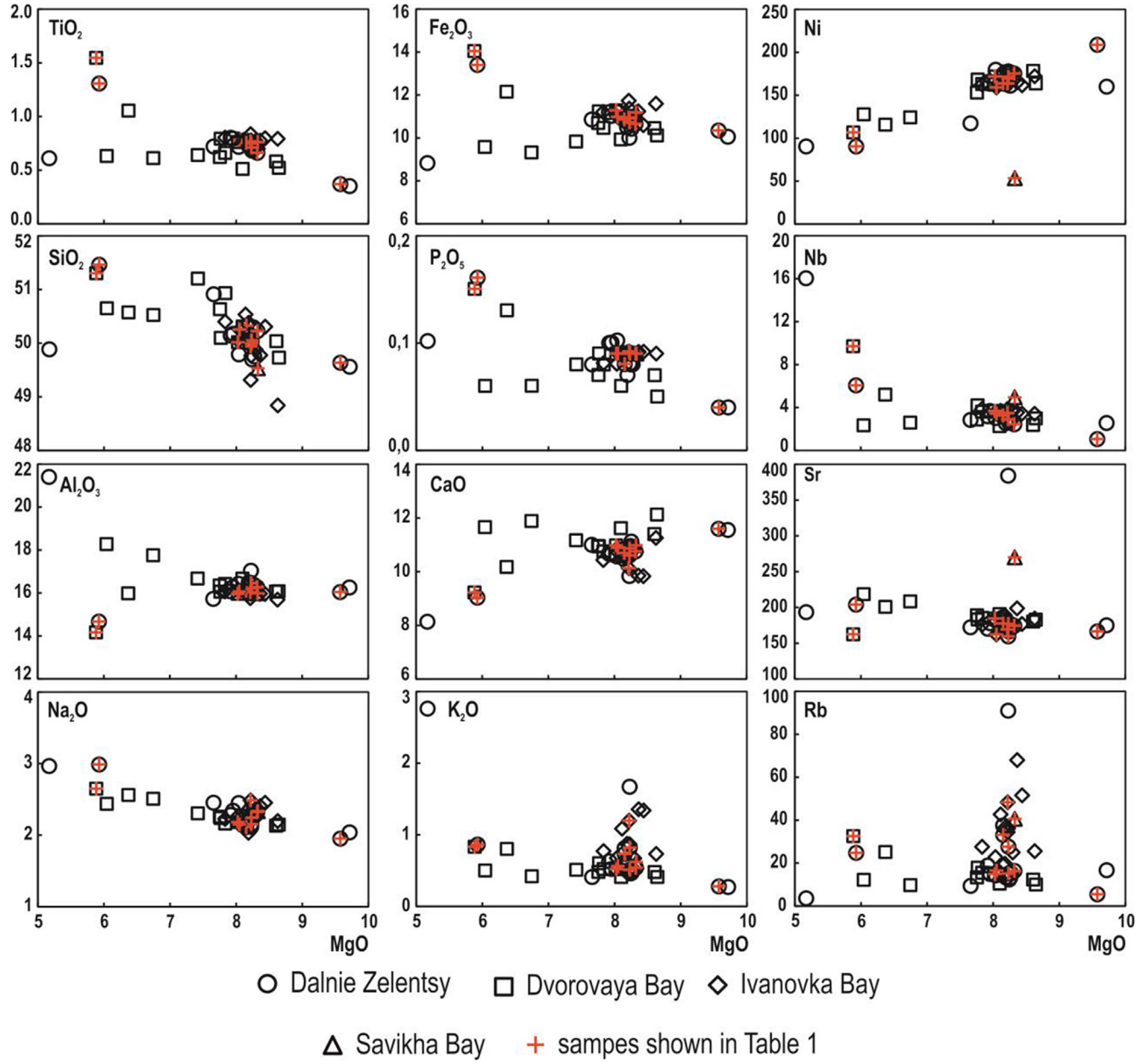


Fig. 3. Variations of major and trace elements vs. MgO in dolerites from studied localities in Murmansk craton. Crosses indicate samples listed in Table 1.

The amphibolitized dolerites mostly preserve their primary composition, but are enriched in K_2O, Rb and Sr and on a binary diagrams they follow a trend significantly different from that typical for magmatic differentiation (Fig. 3).

4.3. Geochronology

U-Pb, Sm-Nd, $^{40}Ar/^{39}Ar$ and Rb-Sr geochronological studies were carried out for samples from thick doleritic sill in Dalnie Zelentsy locality. Three samples from different parts of the sill were selected for investigation: (1) fine-grained dolerite near the bottom (Ca-552-3); (2) medium-grained dolerite from the middle part (Ca-552-6); (3) gabbro-pegmatite from a lense in the upper part of the sill (Ca-552-14). Also, the coarse-grained gabbro-pegmatite from the thick sill of the Dvorovaya locality (sample Ca-676-1) was sampled for the U-Pb baddeleyite dating. The chemical composition of all dated samples is given in Table 1 and shown on Fig. 3.

4.3.1. U-Pb ID-TIMS dating

Baddeleyite from two samples collected from Dalnie Zelentsy and Dvorovaya localities was studied (Fig. 4, Table 2). Approximately 100 grains of transparent or semi-transparent brownish, dark brown elongated, blade-shaped or tabular baddeleyite of good quality were extracted from the sample Ca-552-14 at the Dalnie Zelentsy locality. The optically freshest grains were combined into three fractions comprising 7–8 grains. The two fractions analyzed (# 1 and 2, Table 2) plot on the concordia and yield a well constrained age of 1860 ± 4 Ma (MSWD of concordance = 0.69, the probability of concordance = 0.41) (Fig. 4a). Baddeleyites from analysis #3 are slightly discordant (1.7%) however $^{207}Pb/^{206}Pb$ age for all analyzed fractions is averaged at 1872 ± 2 Ma (MSWD = 2.8) and coincides with the Concordia age.

Sample Ca-676-1 was taken from gabbro-pegmatites in the Dvorovaya locality. Approximately 250 medium brown and dark brown transparent elongated or tabular blades of baddeleyite were recovered from this sample. Three fractions including 3–8 of the most “clean” grains < 50 μm in size were analyzed (Table 2, № 4–6). The data points

Table 1
Major (XRF) and trace element (ICP-MS) composition of representative samples of 1860 Ma dolerite sills in the Murmansk craton.

Sample	Ca-552-3	Ca-552-6	Ca-552-10	Ca-552-14	Ca-562-5	Ca-563-1	Ca-642-1	Ca-649-1	Ca-666-1	Ca-676-1	Ca-676-17
	1	2	3	4	5	6	7	8	9	10	11
wt %											
SiO_2	50.23	50.05	49.64	51.46	50.33	50.00	49.53	50.25	49.94	51.30	50.02
TiO_2	0.66	0.71	0.37	1.31	0.75	0.75	0.76	0.78	0.77	1.55	0.76
Al_2O_3	16.27	16.22	16.05	14.67	16.11	16.47	15.98	16.07	16.04	14.17	15.96
Fe_2O_3	10.65	10.92	10.35	13.40	10.87	10.73	11.19	10.98	10.95	14.05	11.29
MnO	0.16	0.16	0.15	0.18	0.16	0.16	0.18	0.16	0.18	0.20	0.16
MgO	8.32	8.23	9.57	5.93	8.15	8.23	8.33	8.05	8.22	5.88	8.03
CaO	10.76	10.87	11.60	9.04	10.70	10.61	10.98	10.89	10.13	9.23	10.98
Na_2O	2.33	2.25	1.95	2.99	2.11	2.16	2.33	2.15	2.49	2.65	2.18
K_2O	0.54	0.51	0.28	0.87	0.73	0.82	0.62	0.56	1.20	0.83	0.52
P_2O_5	0.09	0.09	0.04	0.16	0.08	0.09	0.09	0.09	0.09	0.15	0.09
LOI	< 0.10	< 0.10	< 0.10	0.37	0.41	0.95	< 0.10	0.31	2.08	0.11	< 0.10
Sum	99.82	99.81	99.84	99.76	99.78	99.76	99.80	99.82	99.80	99.72	99.80
trace elements (ppm)											
Sc	32.9	29.9	29.9	35.9	31.3	30.2	28.4	30.7	32.6	38.8	33.5
V	220	232	164	340	219	211	195	217	228	335	233
Cr	78.3	71.3	104	10.6	71.1	66.0	149	66.6	64.1	13.4	68.4
Co	46.1	47.0	52.2	45.8	48.1	47.6	38.2	44.9	45.3	49.9	47.2
Ni	175	172	209	90.6	164	170	53.4	159	164	107	172
Cu	101	110	68.0	185	109	132	78.5	102	71.5	172	115
Rb	16.4	15.2	5.52	24.8	33.2	27.7	40.6	16.6	48.3	32.6	14.9
Sr	174	177	167	204	178	160	270	162	169	163	185
Y	12.9	14.5	7.81	26.0	14.7	14.4	15.3	14.4	14.5	26.2	14.8
Zr	49.2	63.3	24.4	105	64.5	41.8	71.7	52.4	28.4	106.4	60.9
Nb	2.45	2.90	1.07	6.07	3.27	2.92	4.95	3.45	3.49	9.71	3.69
Cs	0.327	0.221	0.280	0.353	0.625	0.770	0.170	0.372	0.502	0.532	0.246
Ba	170	146	93.8	274	281	243	198	170	188	221	175
La	7.09	8.50	4.01	16.1	8.45	8.00	16.9	8.36	8.88	16.5	8.81
Ce	13.5	16.3	7.37	30.7	15.9	15.0	31.0	17.8	18.5	36.3	19.3
Pr	1.94	2.25	1.11	4.27	2.32	2.26	4.51	2.34	2.23	4.55	2.29
Nd	7.40	8.85	4.36	17.7	9.99	9.16	17.2	9.56	9.17	18.2	10.1
Sm	1.60	2.14	1.05	3.80	1.63	2.22	3.32	2.32	2.37	4.27	2.35
Eu	0.777	0.714	0.552	1.25	0.824	0.861	0.953	0.760	0.759	0.995	0.843
Gd	2.53	2.50	1.27	4.78	2.72	2.59	3.56	2.56	2.42	4.43	2.62
Tb	0.394	0.442	0.275	0.719	0.497	0.423	0.505	0.470	0.405	0.772	0.447
Dy	2.41	2.60	1.45	4.66	2.67	2.81	3.01	2.75	2.73	4.61	2.74
Ho	0.48	0.56	0.29	1.01	0.59	0.57	0.63	0.59	0.61	1.01	0.55
Er	1.61	1.70	1.05	3.05	1.79	1.74	1.89	1.69	1.67	2.86	1.77
Tm	0.224	0.243	0.135	0.453	0.269	0.239	0.268	0.244	0.256	0.411	0.239
Yb	1.51	1.68	0.89	2.85	1.73	1.61	1.68	1.66	1.61	2.81	1.66
Lu	0.230	0.230	0.146	0.442	0.252	0.239	0.268	0.268	0.227	0.444	0.267
Hf	1.54	2.01	0.81	3.27	1.75	1.80	2.23	1.49	0.84	2.89	1.61
Ta	0.172	0.197	0.112	0.364	0.212	0.204	0.354	0.358	0.309	0.662	0.284
Pb	2.17	2.75	1.42	4.79	2.90	2.67	8.32	3.02	3.88	7.20	2.92
Th	1.18	1.52	0.538	2.99	1.50	1.43	3.12	1.65	1.63	4.48	1.61
U	0.305	0.374	0.137	0.738	0.390	0.356	0.821	0.404	0.406	1.074	0.393
La_N/Sm_N	2.76	2.48	2.38	2.65	3.24	2.25	3.17	2.25	2.34	2.40	2.34
Gd_N/Yb_N	1.35	1.20	1.15	1.36	1.27	1.30	1.72	1.25	1.22	1.28	1.28
Eu/Eu*	1.18	0.95	1.47	0.90	1.20	1.10	0.85	0.96	0.97	0.70	1.04
Nb/Nb*	0.31	0.29	0.26	0.32	0.33	0.31	0.25	0.34	0.33	0.41	0.35

1–6 – Dalnie Zelentsy. 1–4 – 30 m thick sill: 1 – fine-grained dolerite from the lower part of the sill; 2 – medium-grained dolerite from the middle part of the sill; 3 – medium-grained melanocratic dolerite from the middle part of the sill; 4 – gabbro-pegmatite from the schlieren in the upper part of the sill; 5–6 – thin sills (< 1 m), fine-grained dolerites. 7 – Savikha locality, fine-grained dolerite from the lower part of the sill; 8 – Ivanovka locality, fine-grained dolerite from the inner contact of the sill; 9 – Drozdovka locality, fine-grained dolerite from the inner contact of the sill; 10–11 – Dvorovaya locality, sill 100 m of thickness: 10 – gabbro-pegmatite from the schlieren from the upper part of the sill; 11 – fine-grained dolerite from the upper part of the sill.

of the baddeleyite were located on the Concordia, giving a reasonably well-constrained Concordia age at of 1863 ± 7 Ma (MSWD of concordance = 0.20, probability of concordance = 0.66) (Fig. 4b).

This U-Pb ID-TIMS baddeleyite study of two samples of dolerites from sills from different localities Ca-552-14 and Ca-676-1 respectively, yields well consistent results at 1860 ± 4 Ma and 1863 ± 7 Ma.

4.3.2. $^{40}Ar/^{39}Ar$ dating

$^{40}Ar/^{39}Ar$ dating of biotite from two samples of dolerites, representing the lower and middle parts of the sill at the Dalnie Zelentsy locality (Ca-552-3 and Ca-552-6, respectively), yields similar results. In sample Ca-552-3, two representative plateaus are obtained, corresponding to the age 1857 ± 20 and 1772 ± 11 Ma (Fig. 4c). In the sample Ca-552-6, one wide plateau corresponds to the age 1865 ± 8 Ma (Fig. 4d, Table 3). The ancient ages present in both samples are indistinguishably close to the U-Pb age of baddeleyite and indicate the time of closure of the K-Ar isotope system for biotite during sill crystallization. The younger $^{40}Ar/^{39}Ar$ age, ca. 1.77 Ga, is apparently associated with a local heating and/or metasomatic event near the bottom (lower contact) of the sill, and coincides with the formation of granites of the Litsa-Araguba complex in the Kola-Norwegian terrain 1.77–1.75 Ga (Vetrin, 2014).

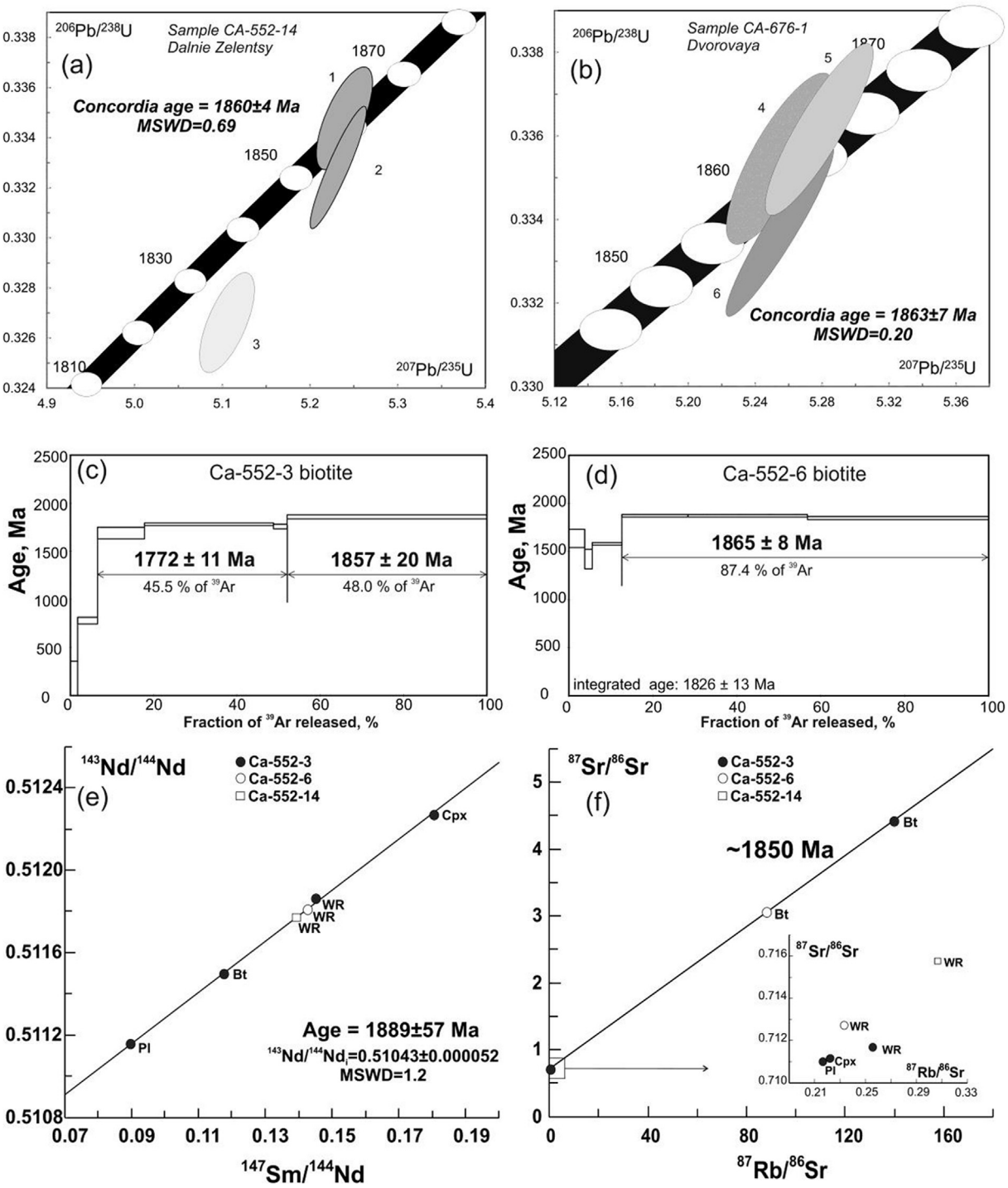


Fig. 4. Geochronology of the dolerite sills. (a, b) – Concordia diagrams showing U-Pb baddeleyite ID-TIMS results for analyses from two samples of the sills. (c, d) – age spectra for two samples of igneous (primary) biotite from the dolerite sill from the Dalnie Zelentsy locality. (e, f) – Sm-Nd (e) and Rb-Sr (f) isochron diagrams for the dolerite sill from the Dalnie Zelentsy locality. WR, whole-rock; Cpx, clinopyroxene; Bt, biotite; Pl, plagioclase.

4.3.3. Sm-Nd and Rb-Sr dating

Three whole-rock samples, Ca-552-3, Ca-552-6 and Ca-552-14, and clinopyroxene, plagioclase and biotite separates from sample Ca-552-3 define an isochron (MSWD = 1.2) with a slope corresponding to an age of 1889 ± 57 Ma, and initial $^{143}Nd/^{144}Nd_i$ = 0.510043 ± 52, ($\varepsilon_{Nd}1860 = -3.00 \pm 0.42$) (Fig. 4e, Table 4). This isochron age within the error is comparable to the U-Pb baddeleyite age of 1860 Ma from gabbro-pegmatite Ca-552-14.

The Rb-Sr isotopic ages of biotite separates $T_{(WR + Bt)}$ of 1848–1859 Ma from two samples of dolerite are also close to the U-Pb age of baddeleyite. Clinopyroxene and plagioclase separates and whole-rock sample of fine-grained Ca-552-3 dolerite on the Rb-Sr isochronous diagram show a strong scatter and do not approximate isochrons (Fig. 4f, Table 4). This may be a result of contamination of dolerites by the hosted granites. This interpretation is well supported by sericite, albite, K-feldspar and epidote aggregates recognized in the central parts of the high-Ca plagioclase (Supplementary, Fig. S2).

Thus, the obtained data indicate that Sm-Nd, Rb-Sr and Ar/Ar isotopic systems in various magmatic minerals in the dolerite sills all close about the crystallization time at 1860 Ma determined by the U-Pb

Table 2
U-Pb ID-TIMS isotopic data for baddeleyite.

#	Characteristics: Size (μm), number of grains, colour and habit	U/Pb*	Pbc/Pbt	Isotopic ratios corrected for blank and common Pb					Rho	Age, Ma		
				$^{206}Pb/^{204}Pb^{**}$	$^{207}Pb/^{206}Pb$	$^{208}Pb/^{206}Pb$	$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$		$^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	$^{207}Pb/^{206}Pb$
Ca-552-14												
1	< 50, 8gr., dark-brown, elongated	3.1	0.001	2171	0.1135 ± 2	0.0103 ± 1	5.2391 ± 188	0.3348 ± 8	0.69	1859 ± 6	1862 ± 6	1856 ± 5
2	< 50, 7gr., dark-brown, elongated, blade-shaped	3.2	0.007	2834	0.1140 ± 2	0.0088 ± 1	5.2321 ± 261	0.3328 ± 15	0.92	1858 ± 9	1852 ± 8	1864 ± 4
3	< 50, 7 gr., brown, tabular and elongated	2.9	0.069	364	0.1134 ± 3	0.0109 ± 2	5.1054 ± 255	0.3266 ± 9	0.66	1837 ± 9	1822 ± 6	1854 ± 7
Ca-676-1												
4	< 50, 3 gr., dark-brown, tabular	3.1	0.001	2100	0.1136 ± 1	0.0025 ± 1	5.2548 ± 105	0.3355 ± 7	0.80	1862 ± 4	1865 ± 4	1858 ± 2
5	< 50, 8 gr., medium-brown, elongated	3.1	0.001	1203	0.1139 ± 1	0.0054 ± 1	5.2781 ± 104	0.3362 ± 6	0.88	1865 ± 4	1868 ± 4	1862 ± 2
6	< 50, 5 gr., dark-brown, tabular	2.9	0.041	1204	0.1142 ± 1	0.0068 ± 1	5.2546 ± 105	0.3337 ± 7	0.94	1862 ± 4	1856 ± 4	1867 ± 1

* – U/Pb ratios for unweighted grains were determined using $^{202}Pb/^{235}U$; Pbc – common Pb; Pbt – total Pb; ** – measured isotopic ratios; Rho – correlation coefficient of $^{207}Pb/^{235}U$ versus $^{206}Pb/^{238}U$ ratios; uncertainties (2σ) refer to last digits of corresponding ratios.

method.

4.4. Rock magnetism

Practically all samples demonstrate thermally stable successive $M_{si}(T)$ curves which can be classified into two types. The majority of samples display thermomagnetic curves with clearly pronounced near-magnetite Curie temperature, T_c (A-type, Fig. 5a). The second type reveals a close to straight line $M_s(T)$ curve shape, combining the magnetite-like and paramagnetic contributions (Fig. 5b, B-type). Hysteresis properties of the samples from A- and B magnetic types also differ: A-type samples display relatively soft ferrimagnetic loops, but the B-type samples look like the sum of the ferro- and paramagnetic phases. The domain state of studied samples was estimated by hysteresis parameters and First Order Reversal Curves (FORCs); the data fall into the pseudo-single-domain (PSD) and multi-domain (MD) range on a Day plot (Fig. 5c), suggesting the mixture of coarse and fine grains. This conclusion has been confirmed by FORCs for the most representing samples (Fig. 5d). Nevertheless, the carriers of NRM are biased to small PSD and single domain (SD) grains and we detect their contribution to NRM by usage of the thermomagnetic criterion (Scherbakov and Scherbakova, 2001), that was applied to all samples used for paleointensity determinations (Fig. 5a, b, on the right). All high-temperature pTRMs showed a pronounced SD or mixed SD-PSD behavior with small pTRM tails, while low-temperature temperature pTRMs tend to demonstrate substantial PSD-MD tails. Remembering that reliable Thellier paleointensity estimations can be obtained namely on SD-PSD particles carrying TRM, this observation supports the validity of studied samples for paleodirection and paleointensity studies, and confirms that only high-temperature pTRMs should be used for this purpose. Temperature dependence of magnetic susceptibility is in general agreement with the results described above and confirms that magnetite or Ti-poor titanomagnetite is the main carrier of magnetization, as evidenced by a sharp drop in susceptibility near the Curie point of magnetite 580 °C (Fig. 5e).

The presence of SD-PSD magnetite particles in rocks is often associated with the deuteric oxidation of coarse titanomagnetite (TM) grains that occurred during cooling of magmatic bodies and resulted in the appearance of magnetite-ilmenite intergrowth structures. This scenario agrees well with the electron microscopic images (Fig. 5f), showing a typical heterophase oxidation structure of initial TM coarse grains characterized by the exsolution of TM into alternate ilmenite and magnetite lamellae. To make the exsolution structure more visible, polished sections were etched with hydrochloric acid which helps to distinguish ilmenite and magnetite lamellae since etching dissolves magnetite but not ilmenite. Typical sizes of magnetite cells vary from submicron to a few μm sizes thus falling in the PSD range.

The electron microscopic observations are supported by results of X-ray powder diffraction performed on magnetic extraction of the samples 322-17 and 505-17 (Ivanovka/Drozdovka localities, sites 22-17 and 37-17, respectively) which reveal a presence of two phases: pure magnetite with a cell parameter a = 0.8394 nm and a rhombohedral ilmenite phase with a = 0.5087 nm. The lines are well shaped with no evidence of another spinel phase or significant line broadening. The corresponding powder X-ray diffraction patterns are shown in Supplementary Materials in Fig. S3. These findings also unambiguously point to the occurrence of magnetite-ilmenite exsolution structure as a result of high-temperature oxidation of initial TM.

4.5. Paleomagnetism

Magnetic treatment and component analysis. A total of 176 samples (294 specimens) from 18 sampling sites were treated: 190 specimens from 16 sites, representing 14 separate outcrops, gave stable results. The rest of samples were unstable or had a maximum angular deviation (MAD) > 6 and were not included in the mean remanence directions.

Table 3
Ar/Ar dating results.

T°C	t (min)	^{40}Ar, 10^{-9} nsm^3	$^{40}Ar/^{39}Ar$	± 1σ	$^{38}Ar/^{39}Ar$	± 1σ	$^{37}Ar/^{39}Ar$	± 1σ	$^{36}Ar/^{39}Ar$	± 1σ	Ca/K	$\Sigma^{39}Ar$ (%)	Age, Ma	± 1σ
Sample Ca-552–3 Biotite (7.11 mg)														
J = 0.003905 ± 0.000040; Integrated age = 1751.0 ± 16.0 Ma														
500	10	3.1	201.1	24.804	0.313	0.24600	61.197	26.753	0.6200	0.1423	220.3	1.7	121.8	233.9
650	10	7.1	169.6	4.085	0.332	0.09458	0.646	7.442	0.1066	0.0227	2.3	6.5	778.2	34.7
775	10	48.8	490.9	21.606	0.066	0.06251	27.965	4.941	0.3189	0.0459	100.7	17.8	1688.1	61.1
875	10	124.7	456.4	1.856	0.024	0.01536	1.257	1.109	0.0871	0.0038	4.5	48.8	1779.7	12.8
960	10	13.0	453.5	6.954	0.044	0.02187	4.347	1.943	0.1067	0.0152	15.6	52.0	1756.6	23.9
1130	10	205.5	485.8	5.855	0.028	0.00224	4.669	1.047	0.0850	0.0120	16.8	100.0	1856.8	20.4
Sample Ca-552–6 Biotite (11.62 mg)														
J = 0.003884 ± 0.000040; Integrated age = 1825.6 ± 4.0 Ma														
500	10	25.3	408.3	28.919	0.150	0.15499	25.341	23.198	0.0961	0.0663	91.2	3.8	1635.2	94.7
625	10	9.9	337.5	25.338	0.084	0.08796	0.829	13.420	0.0985	0.0739	3.0	5.6	1420.6	101.8
800	10	46.1	399.4	2.364	0.087	0.02058	5.767	0.823	0.1334	0.0059	20.8	12.6	1578.2	13.5
925	10	122.2	474.2	1.634	0.021	0.00224	2.713	1.122	0.0171	0.0031	9.8	28.4	1871.9	12.9
1025	10	224.8	482.7	1.596	0.003	0.00164	2.221	1.906	0.0465	0.0025	8.0	56.8	1871.4	12.7
1130	10	331.5	469.2	3.600	0.018	0.00567	3.037	0.915	0.0341	0.0075	10.9	100.0	1846.6	15.9

Results of stepwise thermal magnetic treatment and AF demagnetization indicate the presence of a high-quality paleomagnetic record in most of the studied samples. Natural remanent magnetization (NRM) is represented by one or two components. The least stable component is isolated in the interval 20–180 °C, 0–10 mT and has a large dispersion that is interpreted as a result of viscous or laboratory remagnetization. The second, stable unipolar characteristic component (ChRM) is identified in the interval 20-180-600 °C or 20–50 mT. The ChRM direction for each site, mean directions and corresponding paleomagnetic pole are presented in Table 5 and on Fig. 6 and discussed below.

Sills in the *Dalnie Zelentsy locality* were sampled in 4 sites in which the ChRM is confidently distinguished, and the results of thermal and AF treatment are consistent with each other. The rocks from the large sill in the *Savikha locality*, sampled at site 15–17 (Ca-642), carry a paleomagnetic record of excellent quality: the results of both T and AF magnetic treatments are in good agreement. Due to the small number of samples, the mean paleomagnetic direction for this site was calculated using all single directions of ChRM.

Sills in the *Drozdovka and Ivanovka localities* were sampled in 8 sites. The rocks from site 21–17 (Ca-649) do not carry an interpreted paleomagnetic signal. NRM of the samples from site 23–17 (Ca-652) carry only one component of magnetization, which coincides with well-known Devonian paleomagnetic directions for this region (Veselovskiy et al., 2013). Likely, this NRM component is a result of Devonian remagnetization, because Devonian intrusive bodies are widely represented in this area. Such an explanation is consistent with the results of petrographic and geochemical studies, suggesting significant changes in the mineral and chemical composition of the sills in some of the discussed outcrops. In the remaining sites, the ChRM is confidently distinguished by the results of both types of magnetic cleaning, and is in a good agreement between the sites.

In the *Dvorovaya locality*, a thick (~100 m) sill was sampled at three sites: site 42-17 corresponds to the middle part of the sill, site 43-17 – the upper part, and site 44-17 – the sill margin (endocontact) in the lower part of the sill. All samples carry an excellent paleomagnetic record; however, it is noteworthy that the results of T- and AF-treatment are not in agreement. We suggest that the mentioned feature could be explained by different paleomagnetic stability of the viscous component of NRM to the demagnetizing factor in multi-domain magnetite, which is prevailing in these rocks according to the Day plot (Fig. 5c). Indeed, the paleomagnetic directions, obtained by thermal treatment, lie very close to the present-day field direction and can be a result of contamination of the ChRM by the modern magnetization. For this reason, we do not use the results of T-cleaning when calculating the mean direction of the ChRM for these sites as well as for the total mean.

Site-mean directions of the most stable, characteristic component of magnetization (ChRM) form directional groups, which more or less correspond to the sampling localities (Fig. 6f). It should be noted, that the mean paleomagnetic direction of the ChRM from the Dalnie Zelentsy sills differs significantly from the corresponding directions of the sills from the Ivanovka/Drozdovka and Dvorovaya localities. Assuming that the observed difference of site-mean paleomagnetic directions is a result of the secular variation, this may indicate a close, but slightly different age of magmatism in distant localities. Taking into account the remoteness of the Dalnie Zelentsy locality from other studied areas (~120 km), this assumption looks quite convincing. The preliminary mean ChRM direction of the Savikha sill lies close to the group of site-mean directions of the neighboring localities – Ivanovka/Drozdovka and Dvorovaya.

The total mean paleomagnetic direction of ChRM (Fig. 6f, Table 5) was calculated on a site-mean level. At the same time, if the results of T- and AF-cleaning in the same sampling site did not differ significantly,

Table 4
Sm-Nd and Rb-Sr isotope data for samples from 1860 Ma dolerite sill, Dalnie Zelentsy locality, Murmansk craton.

Sample	Rock/Mineral	Sm, ppm	Nd, ppm	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$	εNd	Rb ppm	Sr ppm	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$	$(^{87}Sr/^{86}Sr)_T$
Ca-552-3	dolerite	2.07	8.64	0.1449	0.511863 ± 10	−2.8	21.1	230	0.265	0.711692 ± 9	0.70459
Ca-552-3	Cpx	4.95	16.6	0.1803	0.512269 ± 14	−3.3	1.89	23.7	0.231	0.711132 ± 51	0.70495
Ca-552-3	Pl	0.493	3.31	0.0899	0.511154 ± 11	−3.5	29.8	381	0.226	0.710997 ± 7	0.70494
Ca-552-3	Bt	0.421	2.17	0.1176	0.511497 ± 11	−3.4	539	11.2	139	4.42169 ± 592	0.70108
Ca-552-6	dolerite	2.40	10.2	0.1423	0.511815 ± 6	−3.1	19.4	229	0.244	0.712745 ± 10	0.70623
Ca-552-6	Bt						522	17.2	88.2	3.06107 ± 39	0.70154
Ca-552-14	gabbro-pegmatite	4.25	18.5	0.1391	0.511783 ± 9	−3.0	28.2	256	0.318	0.715794 ± 10	0.70729

The initial εNd^T and $(^{87}Sr/^{86}Sr)_T$ values were calculated for the emplacement age of the sill T = 1860 Ma.

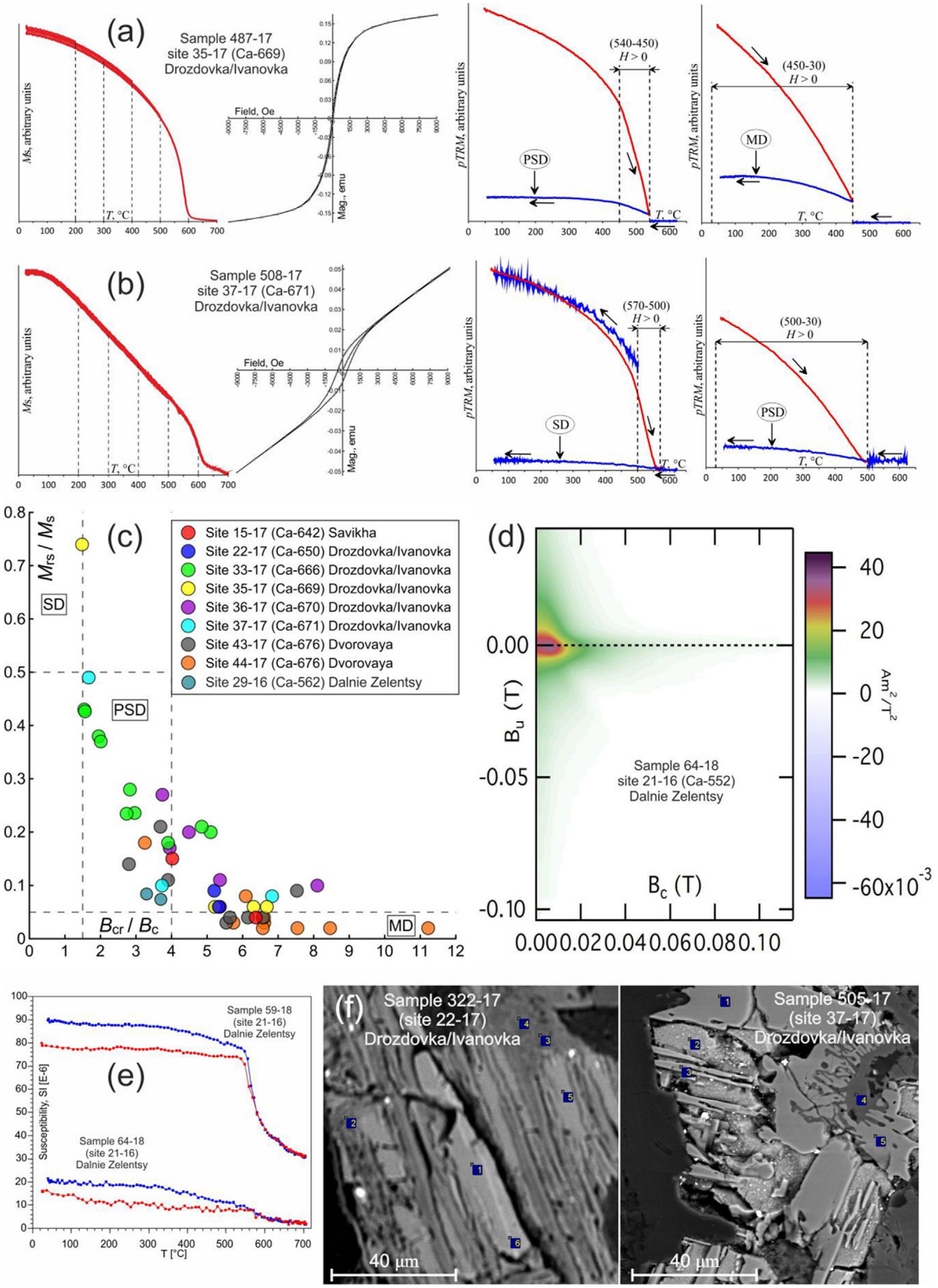


Fig. 5. Rock magnetism. (a, b) – set of $M_{si}(T)$ curves, hysteresis loop and domain structure estimation by the thermomagnetic criterion (see text). Arrows depict heating to T_2 and cooling to T_0 (tail measurement) of pTRMs(T_2, T_1) created in high (left plot) and low (right plot) temperature intervals, and dotted lines mark the T-interval where pTRM was performed; (c, d) – Day plot and typical FORC-diagram showing domain state; (e) – thermal susceptibility dependence for two typical samples of sills; (f) – SEM backscattering image of samples: a) 322-17, b) 505-17 Dark patterns indicate ilmenite grains and lamellae, light ones (etched) correspond to near- magnetite lamellae. On all plots the red color of curves means the heating stage, blue – the cooling stage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
ChRM directions for each sampling site (site-means), total mean direction and corresponding paleomagnetic pole.

NN	Site Geochem.	Site Paleomag	Slat (°N)	Slong (°E)	Treatment	N/n	W	Dec (°)	Inc (°)	K	α95 (°)	Age (Ma)/Comment
Dalnie Zelentsy:												
1	Ca-552	21–16	69.12383	36.0481	t	24/9	0.5	326.5	51.3	37	8.6	1859.5 ± 3.8, Bd, U-Pb
					af	24/7	0.5	331.0	46.9	99	6.1	1865 ± 8 ,Ar/Ar, Biotite
												1848 ± 1.5, Rb-Sr
												1886 ± 58, Sm-Nd
2	Ca-560	28–16	69.10347	36.07004	t	12/10	1.0	342.0	53.9	35	8.3	
3	Ca-562, 563, 564	29–16	69.11309	36.10683	t	12/9	0.5	334.3	49.2	276	3.1	
					af	12/9	0.5	325.4	41.1	313	2.9	
4	Ca-565	32–16	69.11257	36.10349	t	9/4	0.5	339.9	46.2	73	10.8	
					af	9/5	0.5	342.3	46.1	81	8.6	
	Locality mean:		***69.11***	***36.08***	***af + t***	***4 sites***	**–**	***335.2***	***48.8***	***135***	***7.9***	
Savikha Bay:												
5	Ca-642	15–17	68.21873	39.04927	t	3/3	**–**	342.9	64.3	94	12.8	
					af	3/3	**–**	338.1	52.9	113	11.6	
	Locality mean:		***68.22***	***39.05***	***af + t***	***6 samp.***	**–**	***340.1***	***58.6***	***71***	***8.0***	Not reliable
Drozdovka and Ivanovka Bays:												
6	Ca-649	21–17	68.33955	38.45776	t, af	12/0	**–**	–	–			
7	Ca-650	22–17	68.33640	38.45835	t	9/8	1.0	347.8	57.7	163	4.4	
8	Ca-652*	23–17	68.32858	38.49141	t	15/15	**–**	44.6	36.8	172	2.9	Devonian remagnetization
					af	9/9	**–**	42.9	40.8	166	4.0	
9	Ca-666	33-17a	68.32394	38.43556	t	13/10	0.5	7.9	44.0	104	4.7	Lower contact
					af	13/6	0.5	8.9	40.2	38	11.0	
		33-17b	68.32767	38.43676	t	7/5	0.5	356.3	49.3	120	7.0	Upper contact
					af	7/2	0.5	352.3	49.4	–	–	
10	Ca-669	35–17	68.33360	38.44299	t	12/10	0.5	347.8	60.2	158	3.9	
					af	12/10	0.5	345.7	59.0	108	4.7	
11	Ca-670	36–17	68.34026	38.43563	t	8/2	0.5	1.8	54.0	–	–	
					af	8/5	0.5	4.7	47.9	50	10.9	
12	Ca-671	37–17	68.33576	38.43576	t	6/5	0.5	357.2	51.4	71	9.1	
					af	6/6	0.5	1.5	47.0	41	10.6	
13	Ca-671	2-15a	68.34017	38.43663	t	4/4	1.0	0.9	50.0	268	5.6	Inner contact
		2-15b			t	4/4	1.0	330.4	58.2	405	4.6	10 m from the contact
	Locality mean:		***68.34***	***38.44***	***af + t***	***8 sites***	**–**	***355.2***	***52.7***	***73***	***6.5***	
Dvorovaya Bay:												
14	Ca-676	42–17	68.43871	38.20623	t	13/9	**–**	26.3	72.6	113	4.9	1863 ± 7, Bd, U-Pb
					af	13/12	1.0	350.9	60.5	141	3.7	
15	Ca-676	43–17	68.43885	38.21042	t	9/8	**–**	7.1	66.3	64	7.0	
					af	9/8	1.0	353.0	58.3	608	2.2	
16	Ca-676	44–17	68.43440	38.20900	t	12/6	**–**	30.0	72.0	303	3.9	
					af	12/11	1.0	353.3	54.3	126	4.1	
	Locality mean:		***68.44***	***38.21***	***af***	***3 sites***	**–**	***352.5***	***57.7***	***634***	***4.9***	
	Total mean:		**68.54**	**37.88**	**t + af**	**16**		**348.4**	**53.3**	**68**	**4.5**	
Pole:					**Plat = 54.7°; plong = 234.7°; A95 = 5.2°, dp/dm = 4.3°/6.3°; φ_m = 33.9°**							
	Ca-633	7–17	68.64293	37.79435	t, af	9/9		147.4	−77.1	282	3.1	~2680, Bd, U-Pb: Dyke

Note: Slat/Slong – site geographical latitude/longitude. Dec, Inc – declination/inclination. α95/A95 – radius of 95% confidence cone of mean direction/pole. N/n – number of specimens passed magnetic cleanings/used in statistics; each t- or af-treated specimen represents one paleomagnetic sample. W – statistical weight (0.5 or 1.0) that is applied to the corresponding site-mean ChRM direction when the Locality/Total mean is calculating. Absence of W-value means that current ChRM direction has no contribution to the Locality/Total mean. Plat/Plong – paleomagnetic pole latitude/longitude. dp/dm – minimal/maximal semi-axes of 95% confidence ellipse; φ_m – paleolatitude.

then the total mean was calculated using both T and AF results taken with a statistical weight of "0.5" (for example, Table 5, site 29-16). If the results of T and AF treatment were not in agreement (Table 5, sites 42-17, 43-17, 44-17) or ChRM was distinguished only by one type of magnetic cleaning (Table 5, site 22-17), then a corresponding site-mean direction was used for calculation of the total mean paleomagnetic direction with a statistical weight of "1.0".

4.6. Paleointensity

The paleointensity result for each sample (Supplementary, Table S1) was accepted only if it satisfied the following set of selection criteria (Biggin and Paterson, 2014): (1) The linear fit on the Arai-Nagata diagram used for paleointensity determination should contain at least four consecutive data points (N) over a given temperature interval (T1, T2), (T1 < T2); (2) The NRM fraction *f* used in the linear fit represents at least 20% of the total NRM; (3) The difference between initial pTRM acquisition and pTRM check normalized to the total NRM must be less than 5% within the given (T1, T2) interval; (4) Parameters DRAT and CDRAT must be ≤15%; (5) The NRM vector is univectorial through the fit temperature range (T1, T2); (6) The existence of a wide enough temperature interval (not less than the temperature interval used for the linear fit on the Arai-Nagata diagram) over which the curves NRM (T) and TRM(T) are similar (positive outcome of the Wilson method).

Here, DRAT is the absolute value of the maximal deviation of the pTRM check points from their initial values on the interval (T0, T2), normalized to the length of the NRM-TRM fit line in the interval (T1, T2) (Selkin and Tauxe, 2000); CDRAT is the cumulative DRAT, the sum of the deviations (considering their signs) of the pTRM check points from their initial values in the interval (T0, T2) (Kissel and Laj, 2004).

Examples of the Arai-Nagata and Zijderveld plots as well as the Wilson's diagrams are shown in Fig. 7(a, b). In 8 sites a number of satisfied determinations of paleointensity (B_{anc}) changes from 3 to 11 of sister specimens from 1 to 3 unique samples, but we consider every paleointensity estimation from each sister sample as an independent record. The estimates of Banc by Thellier-Coe and Wilson methods are

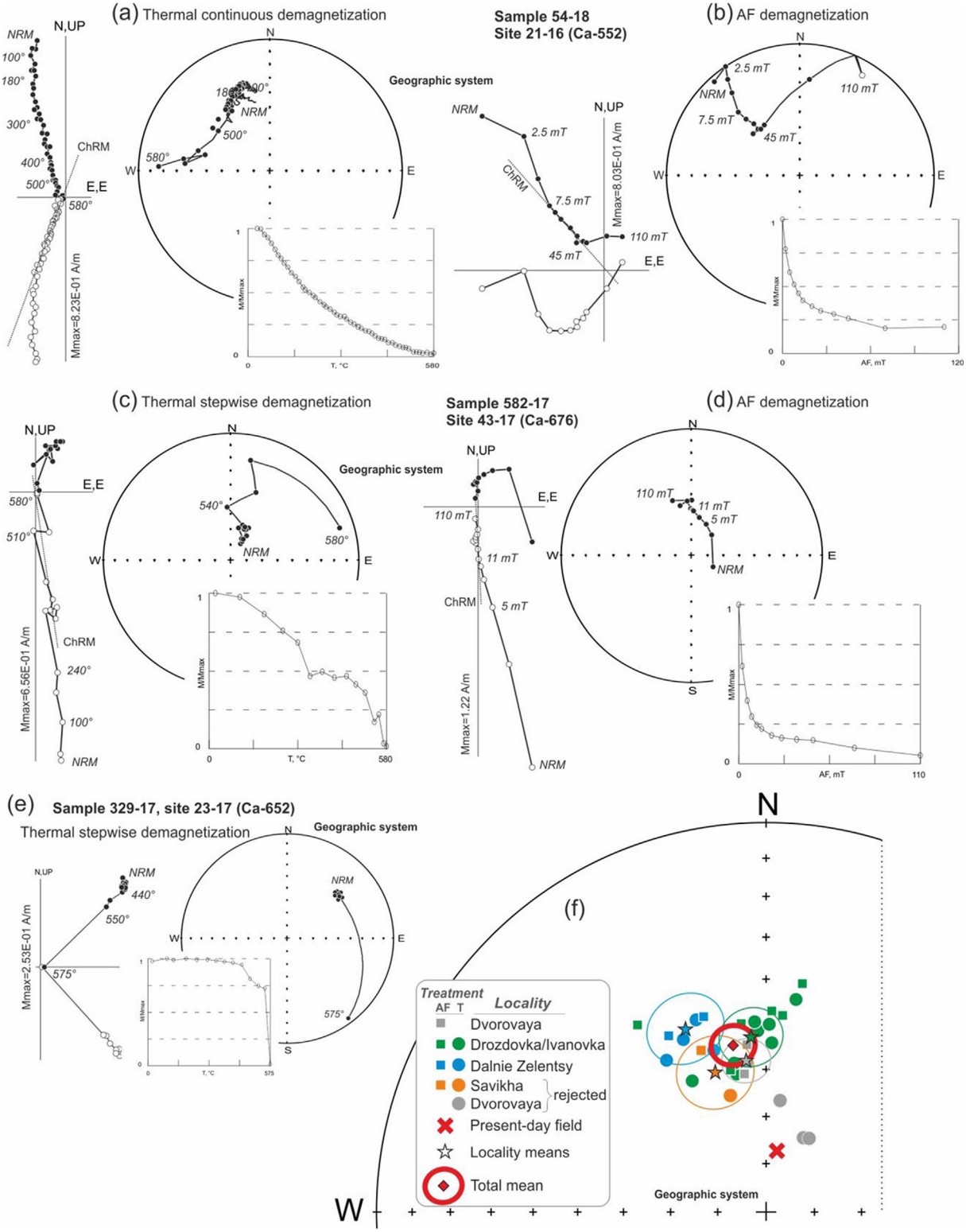


Fig. 6. Representative orthogonal (Zijderveld) demagnetization plots and equal-area stereonet projections (a-e) and the equal-area stereographic projection (f) of all site means reported in this study and the total mean with its circle of 95% confidence (red diamond). In all orthogonal demagnetization plots, closed symbols lie within the horizontal projection and open symbols lie within the vertical projection. In all stereoplots, solid symbols are in the lower hemisphere; open symbols are in the upper hemisphere. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in a good agreement (Table S1), that is also reflected in the similarity between the thermal curves NRM(T) and TRM*(T) on the Wilson diagrams (Fig. 7, a, b). For each site mean values of B_{anc} and the virtual dipole moment (VDM) have been calculated using its paleoinclination *I* (Table 7).

5. Discussion

5.1. The age and the nature of ChRM.

We prove the primary age and thermoremanent (TRM) nature of the

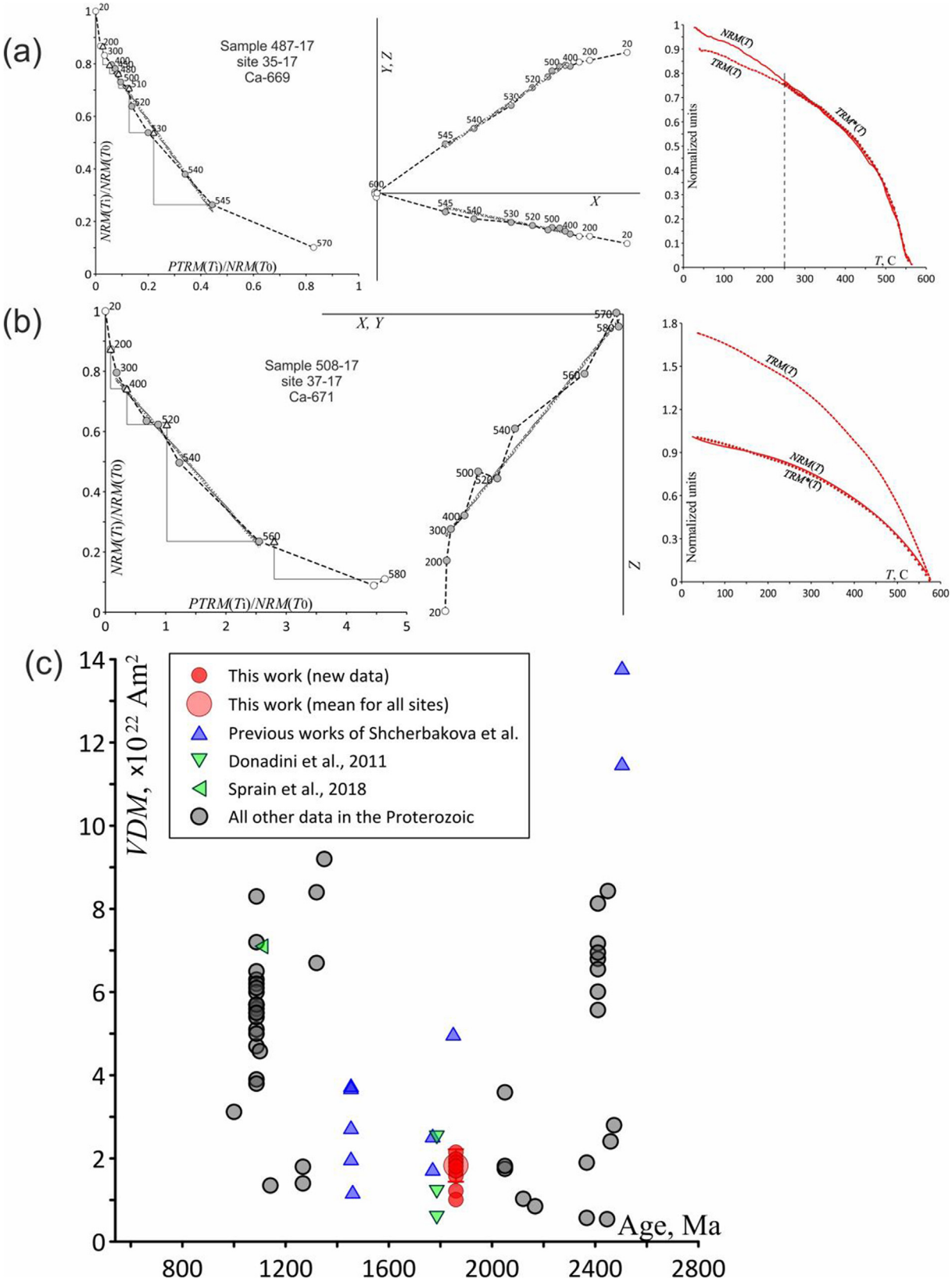


Fig. 7. (a, b) – paleointensity determinations by the Thellier method for two typical samples: Arai-Nagata plot (on the left) and orthogonal plot (in the middle) in sample coordinates; dotted lines mark the T-interval choosing for paleointensity estimation. On the right – paleointensity determinations by the Wilson method. (c) – VDM values obtained in this study (red dots and their mean) in comparison with the selected ones for Proterozoic. The list of references for all data presented in this plot is given in the Supplementary Materials, in the Paleointensity section. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 6
Paleomagnetic poles of Fennoscandia, used in Discussion and on Fig. 8.

Code	Region	Pole	Plat (°N)	Plong (°E)	A95 (°)	Age (Ma)	Qv	References
1	Kola (Pechenga)	Kuetsyarvi formation	24.7	300.8	16.7	2058	4	(Torsvik and Meert, 1995)
2	Karelia	Onega 1975 Ma dykes and sills	44.4	101.5	6.3	1975	5	(Lubnina et al., 2017)
3	Central Kola	Svecofennian remagnetization	54.5	224.0	3.9	1900–1800	3	(Veselovskiy et al., 2017)
4	Svecofennian	Keuruu dykes -C	45.7	230.9	5.1	1870 ± 9	5	(Klein et al., 2016)
5	Murmansk	Eastern Murman Dolerite sills	54.7	234.7	5.2	1860 ± 4	5	*this study*
6	Northern Sweden	Notträsk gabbro	42.6	219.7	3.1	1806 ± 6	5	(Widenfalk et al., 1985)
7	Karelia	Shoksha Formation	39.7	221.1	4.0	1770 ± 10	7	(Pisarevsky and Sokolov, 2001)
8	Central Sweden	Hoting area	43.0	233.3	10.9	1786 ± 10	5	(Elming et al., 2009)
9	Karelia	Ropruchey Sill	39.1	216.6	6.7	1751 ± 3	5	(Lubnina et al., 2012)
10	Northern Sweden	Turinge gabbro-diabase	51.6	220.2	4.8	1703 ± 6	6	(Elming et al., 2018)
11	SW Finland	Åland intrusives -C	23.7	191.4	2.9	1576 ± 3	6	(Salminen et al., 2016)
12	Sweden, Finland	Post-Jotnian Intrusions	1.8	159.1	3.4	1258 ± 12	6	(Elming and Pesonen, 2010)

Note: Plat/Plong – paleomagnetic pole latitude/longitude. A95 – radius of 95% confidence cone. Qv – reliability index (Van der Voo, 1990).

Table 7
Paleointensity data.

Sites	Samples	*N* specimens used for B_{anc}	B_{anc}, μT	B_{anc} st.err., μT	B_{anc} st. dev., μT	I_{anc}, (°)	*VDM*, $\times 10^{22}$ Am^2	*VDM* st. err., $\times 10^{22}$ Am^2	VDM st. dev., $\times 10^{22}$ Am^2	Q_{PI}
15–17	232–235	4	12.4	0.9	1.8	58.6	2.16	0.2	0.3	6
22–17	316–324	5	9.0	0.4	0.8	57.7	1.6	0.1	0.1	7
33–17	440–459	3	9.7	0.6	1.1	44.0	2.0	0.1	0.2	6
35–17	477–496	11	11.1	0.7	2.3	60.2	1.9	0.1	0.4	7
36–17	497–504	4	9.2	0.9	1.8	54.0	1.7	0.2	0.3	5
37–17	505–510	8	6.5	0.5	1.4	51.4	1.23	0.1	0.3	7
43–17	575–583	6	11.1	0.18	0.44	58.3	1.93	0.03	0.07	6
44–17	584–595	5	9.5	1.53	3.4	54.3	1.75	0.3	0.6	6
Mean:			**9.8**	**0.6**	**1.8**	**53.5**	**1.78**	**0.1**	**0.28**	

Note: st. err. – standard error; st. dev. – standard deviation.

characteristic component of the NRM in the studied dolerite sills on the basis of following arguments.

1) The results of U-Pb, Rb-Sr, Sm-Nd and $^{40}Ar/^{39}Ar$ isotopic dating of the Ca-552 sill indicate, that the closure of four independent isotopic systems in different minerals occurred at 1860 Ma. The coincidence (within the standard errors) of the sill crystallization age, determined by the U-Pb age of baddeleyite, and the Ar/Ar age of biotite (closure temperature ~350 °C) indicates the absence of secondary heating of this sill to the temperatures > 350 °C during its life. Nevertheless, it should be noted, that the Ar/Ar dating of biotite of another sample of the same Ca-552 sill indicates an episode of partial loss of radiogenic argon about 1.77 Ga (Fig. 4c). Potentially, the corresponding endogenous activity at ca. 1.77 Ga could be related with the post-orogenic evolution of LKO and could be the cause of complete or partial remagnetization of the basement rocks of the Murmansk craton, including studied dolerite sills. However, comparison of the calculated 1.86 Ga paleomagnetic pole with the key poles of Fennoscandia with an age of ca. 1.77 Ga (Table 6, Fig. 8) shows their significant difference and argues in favor of the primary nature of ChRM in ca. 1.86 Ga sills.
2) We did not succeed in performing at baked contact test in its "classical version" (Butler, 1992): the host rocks – granites and gneisses – are characterized by a weak and/or chaotic paleomagnetic signal, whose principal component analysis (PCA) analysis is not possible. However, the presence of the intrusions older than those studied ca. 1.86 Ga sills within the Murmansk craton with contrasting paleomagnetic record, can be considered as an additional argument in favor of primary nature of the ChRM in the sills. For example, the dyke located on the Barents Sea coast between the Dalnie Zelentsy and Dvorovaya localities – Ca-633 (Table 5) – has a ChRM direction that significantly differs from the total mean direction of the dolerite sills. The isotopic age of this dyke is ~2.68 Ga (ID-TIMS, U-Pb, baddeleyite (Samsonov et al., 2019)).
3) The unipolarity of the ChRM in the studied sills is probably a consequence of their almost equal age, as evidenced by the identity of the petrographic and geochemical characteristics of sills located at a considerable distance (up to 200 km) from each other. In addition, an analysis of global paleomagnetic data at ca. 1.86 Ga suggests that at this time the Earth's magnetic field did not experience an inversion for a long time, possibly about 10–20 million years. This assumption is supported, for example, by the unipolarity of the ChRM from volcano-clastics of Akitkan series of the North-Baikal volcano-plutonic belt, having a thickness of about 6 km and accumulating in the time interval of 1854 ± 5 – 1863 ± 9 Ma (Didenko et al., 2009; Vodovozov et al., 2007).

5.2. VGP or paleomagnetic pole?

So, the obtained data demonstrate that the studied dolerite sills in the Murmansk craton were formed during at single magmatic event. Moreover, there are some indications that the cooling of these sills occurred quickly. In this case, it is necessary to discuss whether secular variation is averaged in the resulting pole or not? According to the current point of view, the averaging of secular variation is achieved in a time span of the order of 10 000 years or more (Tauxe et al., 2009). Below we discuss the pros and cons of the averaging of secular variations in the calculated paleomagnetic pole.

5.2.1. Duration of the sill cooling

In order to obtain a quantitative estimation of the time, elapsed since the formation of dolerite sills to their cooling to the Curie point of magnetite (580 °C), we performed a numerical solution of a three-dimensional thermal model using computer simulation. The model was an extended sill-like body of dolerite with a thickness of 100 m, emplaced into the rocks of the crystalline basement (granites) at a depth of 8 km (the full list of the parameters is available in Supplementary, Paragraph 4). Crystallization depth of melts was estimated using

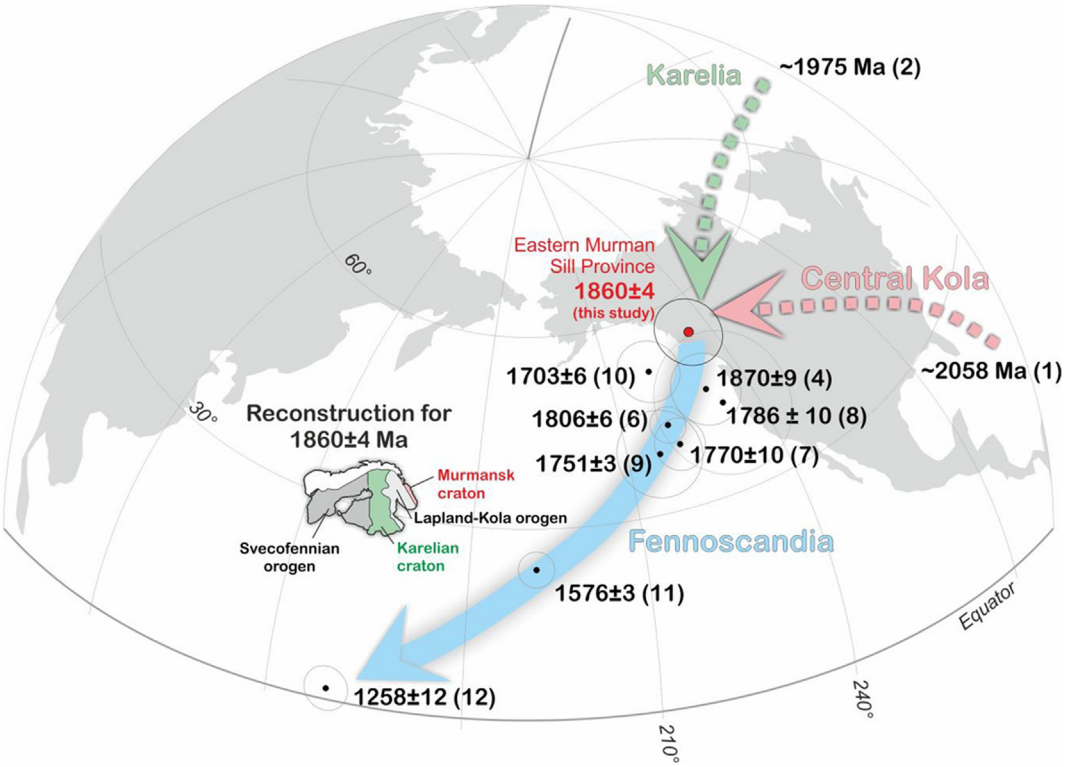


Fig. 8. Comparison of the 1.86 Ga paleomagnetic pole for the Eastern Murman dolerite sills (red circle) with more ancient poles of Karelian craton and Central Kola block (poles #1 and #2), coeval and younger poles of Fennoscandia (poles #4–12). Numbers in brackets correspond to the "Code" column in Table 6. Colored arrows mean the trends of paleomagnetic poles for corresponding units. Reconstructed position of Fennoscandia according to obtained 1.86 Ga paleomagnetic pole is shown also. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

monomineral clinopyroxene geobarometers (Neave and Putirka, 2017; Nimis, 1999; Putirka, 2008). The results (Supplementary, Tables S3 and S4) indicate that crystallization occurred at a pressure < 4.9 kbar. Taking into account that crystallization pressures were estimated by the pyroxene with maximum contents of MgO, i.e. its earliest phase, the obtained pressures correspond to the maximum depths of sill formation. The mean crystallization pressure for pyroxenes from sills of different thickness is 3.7 ± 1.4 kbar, which corresponds to a depth interval of about 10 ± 2 km.

Results of computer modelling show that 100-meter thick dolerite sill reaches a temperature of 580 °C after 2800 years after emplacement (Supplementary, Fig. S4). For instance, its cooling to 500 °C takes about 10 000 years. Thus, if we assume that the emplacement of a group of fast-cooled sills took place during a time interval about, at least, several thousand years, then the secular variations in their paleomagnetic record should be averaged. The presence of directional groups of site-mean ChRM directions (Fig. 6), confirmed by significant difference between the locality-mean ChRM directions of the most distant sites – Dalnie Zelentsy and Dvorovaya (γ/γ_{cr} = 14.1°/8.9°), Dalnie Zelentsy and Ivanovka (γ/γ_{cr} = 13.2°/10.1°), can be observed as an additional evidence of a slightly different magnetization age for the western and eastern sills.

5.2.2. VGP scatter

An additional argument in a favor of the appropriate secular variations averaging at the calculated pole of the sills can be obtained, if we compare the dispersion of the virtual geomagnetic poles (VGPs) for each site relative to the mean pole, with an expected amplitude of the secular variations (S_b) for a given time and latitude. An empirical estimation of S_b values for the Proterozoic and Archean has been performed recently (Smirnov et al., 2011), where the authors used, in particular, the VGPs from Flaherty volcanics (1.87 Ga) and Mashonaland dolerites (1.88 Ga) and have got S_b = 11.6 ± 1.7° and S_b = 14.8 ± 4.0°, respectively. The VGP scatter for the studied sills, estimated using the PmagPy software (Tauxe et al., 2016), gives S = 11.9 ± 2.1°, which is in a good agreement with the model of secular variations for Proterozoic (Smirnov et al., 2011). This is an additional argument in favor of that the pole, calculated for 1.86 Ga sills, is paleomagnetic pole. It's interesting to note, that VGP scatter for paleomagnetic pole, obtained recently for the numerous dykes within the whole Kola Peninsula and interpreted as a result of 1.9–1.8 Ga regional remagnetization (Veselovskiy et al., 2017), is S_b = 12.5 ± 2.3°, which is also in agreement with the mentioned model of secular variations. This result can be considered as an additional argument in support of successful averaging of secular variations both in the magnetic record of 1.86 Ga dolerite sills and remagnetized dykes in the study mentioned above.

5.3. The reliability of obtained paleomagnetic pole

According to the criterion of reliability of paleomagnetic data Qv (Van der Voo, 1990) and its modified version for Precambrian AV (Veikkolainen et al., 2017), the obtained paleomagnetic pole of the 1.86 Ga sills of the Murmansk craton can be considered as key pole and has the ratios AV = 5 (111110) and Qv = 5 (1111100):

1) the age of the sills is very well determined (Qv = 1);
2) the sampling is representative (Qv = 2);
3) the laboratory treatment is good enough (Qv = 3);
4) the age of magnetization is not constrained by field tests due to the lack of reversal polarity of ChRM, an inconclusive contact test because of weak magnetization of host rocks, and absence of sedimentary cover. But as discussed in the text the multidisciplinary data – geochemistry, petrography, geochronology and so on – strongly indicates that ChRM has a primary nature and makes possible to estimate the age of ChRM. Therefore, we believe that this

Van der Voo criterion is satisfied and the age of magnetization is reliably established (Qv = 4);
5) there is a general agreement in the pole position from units of similar age from a broad region, and a structural control has been done (Qv = 5);
6) reversal test is impossible because all the data are of a single polarity;
7) the obtained 1.86 Ga paleomagnetic pole is close to the younger ones.

Thus, the obtained pole can be considered as the first paleomagnetic pole of the Murmansk craton, obtained using modern methodology, and as a 1.86 Ga key paleomagnetic pole of Fennoscandia. This pole also fully meets the key-pole requirements listed in Buchan (2014): the age of the paleopole is precisely determined and the paleopole is of good quality.

5.4. Paleotectonic implications

The available geological, structural, and age data (Daly et al., 2006) suggest that at ca. 1.86 Ga Archean crustal segments of Fennoscandia were assembled into a single continental block by this time. The obtained paleomagnetic pole for ca. 1860 Ma Eastern Murman sills allows to test this suggestion on a quantitative level by comparison with the close-aged and younger paleomagnetic poles of the Karelian craton and the Central Kola and Murmansk provinces. For such comparison, we used some of the most reliable paleomagnetic poles, whose age is in the range 1.87–1.25 Ga (Table 6, Fig. 8).

The mafic magmatism of age 1.86 Ga in the Murmansk craton has no analogues in Fennoscandia, excluding ca.1.87 Ga Keuruu mafic dykes within the Svecofennian province, related to post-collisional extension in the Svecofennian orogen (Klein et al., 2016). In the paleomagnetic database PALEOMAGIA (Veikkolainen et al., 2017), the paleomagnetic data for Fennoscandia with an age of 1.86 Ga are completely absent, so we compared the pole of 1.86 Ga dolerite sills with the 1.87 Ga pole of Keuruu dykes, which is positioned by the authors as a key for Fennoscandia (Klein et al., 2016). The compared poles differ significantly (γ/γ_{cr} = 9.3°/6.9°), although they are located quite close to each other (Fig. 8). We have to note, that the observed difference between two poles could be a result of some unaccounted factors, such as tectonics, but an uncertainty of the age of the Keuruu dykes seems to be the most significant reason because of uncertainty in the origin of dated zircons (Klein et al., 2016).

All other 1.87–1.70 Ga paleomagnetic poles lie in close proximity to each other, that suggests the absence of significant relative movements between the Murmansk and Karelian cratons over the last 1.86 Gyr and indicates the spatial stability of the Nuna's core in the 1.87–1.70 Ga interval. On the contrary, the trend of 1576 and 1258 Ma poles (Fig. 8) can be observed as an evidence for the active drift of Fennoscandia in the Mesoproterozoic. It should be noted, that trends of the pre-1.86 Ga paleomagnetic poles of the Murmansk and Karelian cratons, shown in Fig. 8, are approximate and based on two reliable paleomagnetic poles (Table 6, #1 and #2).

Recently, we assumed the existence of a regional remagnetization of the Kola part of the NE of Fennoscandia ca. 1.9–1.8 Ga on the basis of an extensive paleomagnetic dataset for the Precambrian dykes located both within the Murmansk craton and the Central Kola terrain (Veselovskiy et al., 2017). Comparison of this paleomagnetic pole (Table 6, #3) and the obtained 1.86 Ga pole of the Eastern Murman dolerite sills indicates that they do not formally differ (γ/γ_{cr} = 6.2°/6.9°, class B) and are located in close proximity to each other. This can be considered as the evidence for a close timing of the sill formation and the remagnetization event.

The position of Fennoscandia in Late Paleoproterozoic continental reconstructions (1.88–1.81 Ga) is constrained by a limited number of key paleomagnetic poles derived from magmatic formations in Finland within the Svecofennian accretionary orogen, which development was completed about 1.8 Ga (Lahtinen et al., 2005). It is obvious, that paleomagnetic data from the orogens should be verified by comparing with coeval poles from the cratons. However, the corresponding paleomagnetic poles for the Archean cratons – Karelian and, especially, Murmansk – which compose the core of Fennoscandia, are almost absent. According to our paleotectonic reconstruction of Fennoscandia for 1.86 Ga, made on the basis of the obtained paleomagnetic pole, the Kola Peninsula (research area) was located in the northern hemisphere at a latitude ~34°, while Fennoscandia was rotated relative to its present position ~25° clockwise (Fig. 8). This position of Fennoscandia is in general agreement with the paleocontinental reconstructions of the first stages of the Nuna/Columbia supercontinent formation (Evans and Mitchell, 2011; Pisarevsky et al., 2014; Torsvik and Meert, 1995; Zhao et al., 2004; Meert and Santosh, 2017), according to which the modern Barents Sea coast of the Kola Peninsula was practically at the same latitude as the modern north-east coast of Greenland. Thus, we cannot exclude the probability of the close proximity of these regions of Fennoscandia and Laurentia, which is in good agreement with the well-known paleocontinental reconstructions, based on a correlation of Nagssugtoqidian Orogen of Greenland and the Lapland-Kola orogen of eastern Fennoscandia (Zhao et al., 2002). Formation of studied 1.86 Ga doleritic sills in the Murmansk craton followed the collapse in Lapland-Kola orogen ca. 1.87 Ga (Daly et al., 2001) and, possibly, was associated with thinning and extending of the lithosphere due to delamination of the lithospheric root. Almost coeval (ca. 1.87 Ga) mafic magmatism is described in the Ammassalik Intrusive Complex in the Nagssugtoqidian orogeny, Greenland (Lebrun et al., 2018), suggesting the correlation of the eastern Fennoscandia and Greenland orogens.

5.5. Paleointensity

The values of paleointensity obtained here vary between 6.45 and 12.35 μT, and the corresponding *VDMs* fall in the range (1.22–2.15) × 10^{22} Am^2 (Table 7, Fig. 7c). These data may be slightly overestimated due to the cooling rate dependence of the *TRM* of SD grains (Dodson and McLelland-Brown, 1980; Dunlop and Özdemir, 1997). However, the *NRM* carriers are mostly PSD and/or interacting SD grains and there is evidence that such grains are much less, if at all, prone to cooling rate effects (Biggin et al., 2013). Nevertheless, even without any correction for the cooling rate, the data reported here are rather low in comparison with the modern intensities which have an average $B_{anc} \approx 45$ μT and $VDM \approx 7.8 \times 10^{22}$ Am^2.

The high reliability of data can be also confirmed by using the set of QPI criteria proposed recently by Biggin and Paterson (2014). Indeed, application of this set to our data leads to QPI varying from 5 to 7 out of possible 8 (Table 7). Only LITH criterion of requirement of different lithology (petrology) of samples used for the paleointensity determinations failed for all sites.

Regarding the estimation of the anisotropy of magnetic susceptibility (AMS), which is involved in the QPI calculations, note that for the sites 15–17, 22–17, 33–17 and 37–17 we found a degree of anisotropy P_j = (3–6) %. However, for sites 36–17, 43–17 and 44–17 the values of P_j are much higher – 11–22%. Hence, the VDMs values obtained for these sites require an anisotropy correction which was not done here. For this reason, the values of QPI for the sites 36–17, 43–17 and 44–17 were reduced by one in comparison with other sites.

The available Archean/Proterozoic paleointensity data presented in the World paleointensity databases (http://wwwbrk.adm.yar.ru/palmag/index_e.html and http://earth.liv.ac.uk/pint/) are summarized together including the results obtained in this study (Fig. 7c). The PINT data were selected according to criteria of reliability formulated by Perrin and Shcherbakov (1997): only Thellier-type determinations are considered; at least three samples are used for the *VDM* calculation; the standard error of site averaging of B_{anc} did not exceed 15% (the criterion of internal consistency). In total, 59 Thellier-type

paleointensity determinations were found to pass these selection criteria for the Archean/Proterozoic rocks. Among them, the data by Donadini et al. (2011) (Fig. 7c) may be of a special interest as they were obtained also from the Fennoscandian shield, the Hoting gabbro of more or less similar age 1786 Ma. As is seen, the data reported in this study and by Donadini et al. (2011) are in good agreement.

A prominent feature of the data presented in Fig. 7c is an outburst of high field intensities at the Archean–Proterozoic boundary (Shcherbakova et al., 2017b; Tarduno et al., 2006). At the same time, the interval of 2.4–1.5 Ga, corresponding to the Paleo- and Mesoproterozoic, is marked with a noticeably lower intensity of the paleofield with the mean $VDM = 3.2 \times 10^{22}$ Am^2. This sharp decline in field magnitude led to the suggestion of the Proterozoic dipole low hypothesis (Biggin et al., 2009), which is supported by our new data. However, for the younger units of age 1.1–1.5 Ga, the mean *VDM* value again increases to $\sim 6 \times 10^{22}$ Am^2 (Fig. 7). Taking into account these data, Biggin et al. (2015) argued that the solid inner core formed during this interval in the Mesoproterozoic. However, this statement was criticized by Smirnov et al. (2016), who claimed that many of the Middle Proterozoic data are biased to high values due to the so-called MD effect (Scherbakov and Scherbakova, 2001). If this criticism is justified, it means that the onset of the inner core must be delayed until the Early Proterozoic or even Early Phanerozoic, as it was suggested by Shcherbakova et al. (2017a). However, it should be mentioned that Sprain et al. (2018) reported new paleointensity determinations obtained on rocks of ca. 1100 Ma from North American Midcontinent Rift with mean VDM $\sim 5.6 \times 10^{22}$ Am^2, thus providing additional evidence to support the higher Mesoproterozoic field.

Despite all these uncertainties, it seems likely that the current Precambrian *VDM* data suggest non-monotonic paleointensity variations rather than a monotonic increase of intensity which is expected if the inner core emerged before the Late Proterozoic. Theoretical justification for non-monotonic variations of the paleointensity in the geological past was recently demonstrated by Driscoll (2016), where it was shown that under certain conditions that numerical geodynamo models forecast the existence of transitions from the low- to high-intensity field regimes and back as well as from an irregular multipolar structure to the common stable bipolar geometry.

6. Conclusions

This paper presents a comprehensive study of the Paleoproterozoic mafic sills – Eastern Murman Sill Province – in the Murmansk craton, northern Fennoscandia. The main conclusions are:

1) petrographic and geochemical characteristics suggests that studied sills were formed during a single magmatic event and well preserved;
2) U-Pb ID-TIMS ages of baddeleyite give the crystallization age of the sills as at 1860 Ma;
3) the coincidence of Sm-Nd, Rb-Sr and Ar/Ar ages to U-Pb age for one of the studied sills points to quick cooling and the absence of secondary heating of rocks above 350 °C;
4) the age of the characteristic component of magnetization is well-defined by isotopic ages of different minerals using different isotope systems. The primary origin and thermoremanent nature of the ChRM are also supported by rock magnetism and by the presence of contrasting paleomagnetic direction in the older dyke;
5) the obtained paleomagnetic pole has the high scores AV = 5 (Veikkolainen et al., 2017), Qv = 5 (Van der Voo, 1990) and matches to all the criteria applied for Precambrian key-poles (Buchan, 2014). Thus, it can be considered as the first reliable paleomagnetic pole of the Murmansk craton and the 1.86 Ga key-pole of Fennoscandia;
6) the paleotectonic reconstruction of Fennoscandia, corresponding to the presented 1.86 Ga paleomagnetic pole, determines its place in the Nuna supercontinent core at 34° of latitude and suggests the proximity of the northern part of Fennoscandia to the northern part of Greenland (in present-day coordinates). At that time Fennoscandia was rotated ~25° clockwise relative to its present orientation;
7) a set of new 1.86 Ga paleointensity determinations satisfying the modern reliability criteria are reported. The data indicate rather low VDM values ranging from 1.22 to 2.15 $\times 10^{22}$ Am^2 supporting the Proterozoic dipole low hypothesis.

Acknowledgements

This work was supported by Russian Scientific Foundation, grant #16-17-10260. Paleointensity studies were supported by the Ministry of Education and Science (grant MD-1116.2018.5), by the RFBR (grant 19-05-00433) and by the state assignment of GO “Borok” IPE RAS; rock magnetic measurements were supported by the Russian Foundation for Basic Research (grant 17-05-01121) and by the Ministry of Education and Science (grant 14.Z50.31.0017). Ar/Ar studies were partly supported by the Ministry of Education and Science (grant 14.Y26.31.0012). E.B. Salnikova acknowledges support from the state assignment of IPGG RAS (0153-2019-0005). Computer modelling was carried out using the equipment of the shared research facilities of HPC computing resources at Lomonosov Moscow State University.

The authors greatly appreciate the thoughtful comments and suggestions of Richard Ernst and two anonymous reviewers. Our special thanks for help in the field works is addressed to the crew of the “Udacha” boat and to the colleagues from Murmansk Marine Biological Institute, Kola Scientific Center of the Russian Academy of Sciences. Paleomagnetic team thanks Vladimir Tselmovich (GO “Borok”) for invaluable help with rock magnetic studies.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.precamres.2019.01.017.

References

Arzamastsev, A.A., Fedotov, Z.A., Arzamastzeva, L.V., 2009. Dyke magmatism in the Northeastern part of the Baltic Shield. Nauka, St. Petersburg (in Russian).
Aubert, J., Labrosse, S., Poitou, C., 2009. Modelling the palaeo-evolution of the geodynamo. Geophys. J. Int. https://doi.org/10.1111/j.1365-246X.2009.04361.x.
Balagansky, V.V., Glaznev, V.N., Osipenko, L.G., 1998. Early Proterozoic evolution of the northeastern Baltic Shield: a terrane analysis. Geotectonics 2, 16–28 (in Russian).
Bibikova, E., Skiöld, T., Bogdanova, S., Gorbatschev, R., Slabunov, A., 2001. Titanite-rutile thermochronometry across the boundary between the Archaean Craton in Karelia and the Belomorian Mobile Belt, eastern Baltic Shield. Precambrian Res. https://doi.org/10.1016/S0301-9268(00)00117-0.
Biggin, A.J., Badejo, S., Muxworthy, A.R., Dekkers, M.J., 2013. The effect of cooling rate on the intensity of thermoremanent magnetization (TRM) acquired by assemblages of pseudo-single domain, multidomain and interacting single-domain grains. Geophys. J. Int. 193, 1239–1249. https://doi.org/10.1093/gji/ggt078.
Biggin, A.J., Paterson, G.A., 2014. A new set of qualitative reliability criteria to aid inferences on palaeomagnetic dipole moment variations through geological time. Front. Earth Sci. 2, 1–9. https://doi.org/10.3389/feart.2014.00024.
Biggin, A.J., Strik, G.H.M.A., Langereis, C.G., 2009. The intensity of the geomagnetic field in the late-Archaean: new measurements and an analysis of the updated IAGA palaeointensity database. Earth, Planets Sp. https://doi.org/10.1186/BF03352881.
Biggin, A.J., Piispa, E.J., Pesonen, L.J., Holme, R., Paterson, G.A., Veikkolainen, T., Tauxe, L., 2015. Palaeomagnetic field intensity variations suggest Mesoproterozoic inner-core nucleation. Nature. https://doi.org/10.1038/nature15523.
Bogdanova, S.V., 1993. The three-segment hypothesis for the East European Craton. Terra Nov. 5, 313–314.
Bogdanova, S., Gorbatschev, R., Skridlaite, G., Soesoo, A., Taran, L., Kurlovich, D., 2015. Trans-baltic palaeoproterozoic correlations towards the reconstruction of supercontinent Columbia/Nuna. Precambrian Res. 259, 5–33. https://doi.org/10.1016/j.precamres.2014.11.023.
Buchan, K.L., 2014. Reprint of “Key paleomagnetic poles and their use in Proterozoic continent and supercontinent reconstructions: a review”. Precambrian Res. 244, 5–22. https://doi.org/10.1016/j.precamres.2014.01.010.
Butler, R.F., 1992. Paleomagnetism: Magnetic Domains to Geologic Terranes. Blackwell Scientific Publication, Boston, pp. 319.
Chadima, M., Hrouda, F., 2006. Remasoft 3.0: a user-friendly paleomagnetic data browser

and analyzer. Travaux Geophys. XXVII, 20–21.
Coe, R.S., 1967. The determination of paleointensities of the Earth's magnetic field with special emphasize on mechanisms which could cause nonideal behavior in Thellier's method. J. Geomagn. Geoelec. 19, 157–178.
Daly, J.S., Balagansky, V.V., Timmerman, M.J., Whitehouse, M.J., de Jong, K., Guise, P., Bogdanova, S., Gorbatschev, R., Bridgwater, D., 2001. Ion microprobe U-Pb zircon geochronology and isotopic evidence for a trans-crustal suture in the Lapland-Kola Orogen, northern Fennoscandian Shield. Precambrian Res. https://doi.org/10.1016/S0301-9268(00)00116-9.
Daly, J.S., Balagansky, V.V., Timmerman, M.J., Whitehouse, M.J., 2006. The Lapland-Kola orogen: palaeoproterozoic collision and accretion of the northern Fennoscandian lithosphere. Geol. Soc. London, Mem. 32, 579–598. https://doi.org/10.1144/GSL.MEM.2006.032.01.35.
Day, R., Fuller, M., Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain-size and compositional dependence. Phys. Earth Planet. Inter. 13, 260–267. https://doi.org/10.1016/0031-9201(77)90108-X.
Didenko, A.N., Vodovozov, V.Y., Pisarevsky, S.A., Gladkochub, D.P., Donskaya, T.V., Mazukabzov, A.M., Stanevich, A.M., Bibikova, E.V., Kirnozova, T.I., 2009. Palaeomagnetism and U-Pb dates of the Palaeoproterozoic Akitkan Group (South Siberia) and implications for pre-Neoproterozoic tectonics. Geol. Soc. London Spec. Publ. 323, 145–163. https://doi.org/10.1144/SP323.7.
Dodson, M.H., McLelland-Brown, E., 1980. Magnetic blocking temperatures of single-domain grains during slow cooling. J. Geophys. Res. https://doi.org/10.1029/JB085iB05p02625.
Donadini, F., Elming, S.Å., Tauxe, L., Hålenius, U., 2011. Paleointensity determination on a 1.786Ga old gabbro from Hoting, Central Sweden. Earth Planet. Sci. Lett. https://doi.org/10.1016/j.epsl.2011.07.005.
Driscoll, P.E., 2016. Simulating 2 Ga of geodynamo history. Geophys. Res. Lett. https://doi.org/10.1002/2016GL068858.
Dunlop, D.J., Özdemir, Ö., 1997. In: Rock Magnetism: Fundamentals and Frontiers. Cambridge University Press. https://doi.org/10.1017/CBO9780511612794.
Elming, S.Å., Moakhar, M.O., Layer, P., Donadini, F., 2009. Uplift deduced from remanent magnetization of a proterozoic basic dyke and the baked country rock in the hoting area, central Sweden: a palaeomagnetic and 40Ar/39Ar study. J. Int. Geophys. https://doi.org/10.1111/j.1365-246X.2009.04265.x.
Elming, S.Å., Layer, P., Söderlund, U., 2018. Cooling history and age of magnetization of a deep intrusion: a new 1.7 Ga key pole and Svecofennian-post Svecofennian APWP for Baltica. Precambrian Res. https://doi.org/10.1016/j.precamres.2018.05.022.
Elming, S.-A., Pesonen, L., 2010. Recent Developments in Paleomagnetism and Geomagnetism. Eos (Washington. DC) 90, 502.
Enkin, R.J., 1994. A computer program package for analysis and presentation of paleomagnetic data, Pacific Geoscience Centre, Geological Survey of Canada.
Evans, D.A.D., Mitchell, R.N., 2011. Assembly and breakup of the core of Paleoproterozoic-Mesoproterozoic supercontinent Nuna. Geology 39, 443–446. https://doi.org/10.1130/G31654.1.
Fedotov, Z.A., Bayanova, T.B., Serov, P.A., 2012. Spatiotemporal relationships of dike magmatism in the Kola region, the Fennoscandian Shield. Geotectonics 46, 412–426. https://doi.org/10.1134/S0016852112060039.
Glebovitsky, V.A., 2005. Early Precambrian of the Baltic Shield. Nauka, St. Petersburg (in Russian).
Gražulis, S., Daškevič, A., Merkys, A., Chateigner, D., Lutterotti, L., Quirós, M., Serebryanaya, N.R., Moeck, P., Downs, R.T., Le Bail, A., 2012. Crystallography Open Database (COD): an open-access collection of crystal structures and platform for world-wide collaboration. Nucl. Acids Res. https://doi.org/10.1093/nar/gkr900.
Gubbins, D., Alfè, D., Masters, G., Price, G.D., Gillan, M., 2004. Gross thermodynamics of two-component core convection. J. Int. Geophys. https://doi.org/10.1111/j.1365-246X.2004.02219.x.
Hanski, E.J., 2013. Evolution of the Palaeoproterozoic (2.50-1.95 Ga) non-orogenic magmatism in the eastern part of the Fennoscandian shield. Front. Earth Sci. https://doi.org/10.1007/978-3-642-29682-6_6.
Holness, M.B., Cesare, B., Sawyer, E.W., 2011. Melted rocks under the microscope: microstructures and their interpretation. Elements. https://doi.org/10.2113/gselements.7.4.247.
Hölttä, P., Balagansky, V., Garde, A., Mertanen, S., Peltonen, P., Slabunov, A.I., Sorjonen-Ward, P., Whitehouse, M.J., 2008. Archean of Greenland and Fennoscandia. Episodes. https://doi.org/10.1186/1472-6963-9-60.
Jacobsen, S.B., Wasserburg, G.J., 1984. Sm - Nd isotopic evolution of chondrites and achondrites II. Earth Planet. Sci Lett. https://doi.org/10.1016/0012-821X(84)90109-2.
Kaur, P., Zeh, A., Chaudhri, N., Gerdes, A., Okrusch, M., 2013. Nature of magmatism and sedimentation at a Columbia active margin: Insights from combined U-Pb and Lu-Hf isotope data of detrital zircons from NW India. Gondwana Res. https://doi.org/10.1016/j.gr.2012.07.008.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. J. R. Astron. Soc. Geophys. https://doi.org/10.1111/j.1365-246X.1980.tb02601.x.
Kissel, C., Laj, C., 2004. Improvements in procedure and paleointensity selection criteria (PICRIT-03) for Thellier and Thellier determinations: application to Hawaiian basaltic long cores. Earth Planet. Inter. Phys. https://doi.org/10.1016/j.pepi.2004.06.010.
Klein, R., Pesonen, L.J., Mänttäri, I., Heinonen, J.S., 2016. A late Paleoproterozoic key pole for the Fennoscandian Shield: a paleomagnetic study of the Keuruu diabase dykes, Central Finland. Precambrian Res. 286, 379–397. https://doi.org/10.1016/j.precamres.2016.10.013.
Koistinen, T., Stephens, M., Bogatchev, V., Nordgulen, Ø., Wennerstrøom, M., Korhonen, J., 2001. Geological map of the Fennoscandian shield, Scale 1: 2.000.000. Geol. Surv. Finland, Norway, Sweden North-West Dep. Nat. Resour. Russ. https://doi.org/10.1073/pnas.96.11.5897.
Korja, A., Lahtinen, R., Nironen, M., 2006. The Svecofennian orogen: a collage of microcontinents and island arcs. Geol. Soc. London, Mem. https://doi.org/10.1144/GSL.MEM.2006.032.01.34.
Kozlov, N.E., Sorokhtin, N.O., Glaznev, V.N., Kozlova, N.E., Ivanov, A.A., Kudryashov, N.M., Martynov, E.V., Tyuremnov, V.A., Matiushkin, A.V., Osipenko, L.G., 2006. Archaean Geology of the Baltic Shield. Nauka Publ, Saint-Petersburg (in Russian).
Lahtinen, R., Korja, A., Nironen, M., 2005. Chapter 11 Paleoproterozoic tectonic evolution. Precambrian Geol. Dev. https://doi.org/10.1016/S0166-2635(05)80012-X.
Lahtinen, R., Garde, A., Melezhik, V., 2008. Paleoproterozoic evolution of Fennoscandia and Greenland. Episodes 31 (1).
Lahtinen, R., Korja, A., Nironen, M., Heikkinen, P., 2009. Palaeoproterozoic accretionary processes in Fennoscandia. Geol. Soc. London, Spec. Publ. https://doi.org/10.1144/SP318.8.
Larionova, Y.O., Samsonov, A.V., Shatagin, K.N., 2007. Sources of Archean sanukitoids (high-Mg subalkaline granitoids) in the Karelian craton: Sm–Nd and Rb–Sr isotopic-geochemical evidence. Petrology 15, 530–550.
Lebrun, E., Arting, T.B., Kolb, J., Fiorentini, M., Kokfelt, T., Johannesen, A.B., Maas, R., Thebaud, N., Martin, L.A.J., Murphy, R.C., 2018. Genesis of the Paleoproterozoic Ammassalik Intrusive Complex, south-east Greenland. Precambrian Res. 315, 19–44. https://doi.org/10.1016/j.precamres.2018.06.016.
Li, Z.X., Bogdanova, S.V., Collins, A.S., Davidson, A., De Waele, B., Ernst, R.E., Fitzsimons, I.C.W., Fuck, R.A., Gladkochub, D.P., Jacobs, J., Karlstrom, K.E., Lu, S., Natapov, L.M., Pease, V., Pisarevsky, S.A., Thrane, K., Vernikovsky, V., 2008. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambrian Res. 160, 179–210. https://doi.org/10.1016/j.precamres.2007.04.021.
Lubnina, N.V., Pisarevsky, S.A., Soderlund, U., Nilsson, M., Sokolov, S.J., Khramov, A.N., Iosifidi, A.C., Ernst, R., Romanovskaya, M.A., Pisakin, B.N., 2012. New palaeomagnetic and geochronological data from the Ropruchey sill (Karelia, Russia): Implications for late Palaeoproterozoic palaeogeography. In: Mertanen (Ed.), Abstracts Sup. Cont. Symp. pp. 81–82.
Lubnina, N.V., Pisarevsky, S.A., Stepanova, A.V., Bogdanova, S.V., Sokolov, S.J., 2017. Fennoscandia before Nuna/Columbia: paleomagnetism of 1.98–1.96 Ga mafic rocks of the Karelian craton and paleogeographic implications. Precambrian Res. https://doi.org/10.1016/j.precamres.2017.01.011.
Ludwig, K.R., 2012. User's Manual for a geochronological toolkit for Microsoft Excel. Berkeley Geochronol. Cent. https://doi.org/10.1016/j.pneurobio.2014.01.001.
Ludwig, K.R., 1991. PbDat 1.21 for MS-dos: A computer program for IBM-PC Compatibles for processing raw Pb-U-Th isotope data. Version 1.07, US Geological Survey Open-File Report.
Meert, J.G., Santosh, M., 2017. The Columbia supercontinent revisited. Gondwana Res. https://doi.org/10.1016/j.gr.2017.04.011.
Mitchell, R.N., Hoffman, P.F., Evans, D.A.D., 2010. Coronation loop resurrected: oscillatory apparent polar wander of Orosirian (2.05-1.8 Ga) paleomagnetic poles from Slave craton. Precambrian Res. 179, 121–134. https://doi.org/10.1016/j.precamres.2010.02.018.
Nance, R.D., Murphy, J.B., Santosh, M., 2014. The supercontinent cycle: a retrospective essay. Gondwana Res. 25, 4–29. https://doi.org/10.1016/j.gr.2012.12.026.
Neave, D.A., Putirka, K.D., 2017. A new clinopyroxene-liquid barometer, and implications for magma storage pressures under Icelandic rift zones. Am. Mineral. https://doi.org/10.2138/am-2017-5968.
Nimis, P., 1999. Clinopyroxene geobarometry of magmatic rocks. Part 2. Structural geobarometers for basic to acid, tholeiitic and mildly alkaline magmatic systems. Mineral. Petrol. https://doi.org/10.1007/s004100050498.
Olson, P., Deguen, R., Hinnov, L.A., Zhong, S., 2013. Controls on geomagnetic reversals and core evolution by mantle convection in the Phanerozoic. Phys. Earth Planet. Inter. 214, 87–103. https://doi.org/10.1016/j.pepi.2012.10.003.
Perrin, M., Shcherbakov, V., 1997. Paleointensity of the Earth' Magnetic Field for the past 400 Ma: Evidence for a dipole structure during the Mesozoic Low. J. Geomagn. Geoelectr. https://doi.org/10.5636/jgg.49.601.
Piper, J.D.A., 2013. Continental velocity through Precambrian times: the link to magmatism, crustal accretion and episodes of global cooling. Front. Geosci. https://doi.org/10.1016/j.gsf.2012.05.008.
Pisarevsky, S.A., Elming, S.Å., Pesonen, L.J., Li, Z.X., 2014. Mesoproterozoic paleogeography: supercontinent and beyond. Precambrian Res. 244, 207–225. https://doi.org/10.1016/j.precamres.2013.05.014.
Pisarevsky, S.A., Sokolov, S.J., 2001. The magnetostratigraphy and a 1780 Ma palaeomagnetic pole from the red sandstones of the Vazhinka River section, Karelia, Russia. Geophys. J. Int. 146, 531–538. https://doi.org/10.1046/j.0956-540X.2001.01479.x.
Prévot, M., Mankinen, E.A., Coe, R.S., Grommé, C.S., 1985. The Steens Mountain (Oregon) geomagnetic polarity transition: 2. Field intensity variations and discussion of reversal models. J. Geophys. Res. 90, 10417. https://doi.org/10.1029/JB090iB12p10417.
Puchtel, I.S., Hofmann, A.W., Amelin, Y.V., Garbe-Schönberg, C.D., Samsonov, A.V., Shchipansky, A.A., 1999. Combined mantle plume-island arc model for the formation of the 2.9 Ga Sumozero-Kenozero greenstone belt, SE Baltic Shield: isotope and trace element constraints. Geochim. Cosmochim. Acta. https://doi.org/10.1016/S0016-7037(99)00111-8.
Putirka, K.D., 2008. Thermometers and barometers for volcanic systems. Rev. Mineral. Geochem. https://doi.org/10.2138/rmg.2008.69.3.
Roberts, N.M.W., 2013. The boring billion? - Lid tectonics, continental growth and environmental change associated with the Columbia supercontinent. Geosci. Front. 4, 681–691. https://doi.org/10.1016/j.gsf.2013.05.004.
Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic Supercontinent. Gondwana Res. 5, 5–22. https://doi.org/10.1016/S1342-937X(05)

70883-2.
Rogers, J.J.W., Santosh, M., 2009. Tectonics and surface effects of the supercontinent Columbia. Gondwana Res. https://doi.org/10.1016/j.gr.2008.06.008.
Salminen, J.M., Klein, R., Mertanen, S., Pesonen, L.J., Fröjdö, S., Mänttäri, I., Eklund, O., 2016. Palaeomagnetism and U-Pb geochronology of c. 1570 Ma intrusives from Åland archipelago, SW Finland - implications for Nuna. Geol. Soc. Spec. Publ. https://doi.org/10.1144/SP424.3.
Samsonov, A.V., Stepanova, A.V., Salnikova, E.B., Arzamastsev, A.A., Veselovskiy, R.V., Larionova, Yu.O., Larionov, A.N., Egorova, S.V., Stifeeva, M.V., 2019. Neoarchean "Pechenga-type" mantle plume in the northern Fennoscandia: 2680 Ma mafic dykes in the Murmansk Craton. In: Proceedings of LI Tectonic conference "Issues of tectonics of continents and oceans". GEOS, Moscow, pp. 216–220 (in Russian).
Scherbakov, V.P., Scherbakova, V.V., 2001. On the suitability of the Thellier method of palaeointensity determinations on pseudo-single-domain and multidomain grains. J. Int. Geophys. https://doi.org/10.1046/j.0956-540X.2001.01421.x.
Selkin, P.A., Tauxe, L., 2000. Long-term variations in palaeointensity. Philos. Trans. R. Soc. A Math. Phys. Eng. Sci. https://doi.org/10.1098/rsta.2000.0574.
Shcherbakova, V.V., Shcherbakov, V.P., Heider, F., 2000. Properties of partial thermoremanent magnetization in pseudosingle domain and multidomain magnetite grains. J. Geophys. Res. Solid Earth. https://doi.org/10.1029/1999JB900235.
Shcherbakova, V.V., Lubnina, N.V., Shcherbakov, V.P., Zhidkov, G.V., Tsel'movich, V.A., 2017b. Paleointensity determination on Neoarchaean dikes within the Vodlozerskii terrane of the Karelian craton. Izv. Phys. Solid Earth. https://doi.org/10.1134/S1069351317050111.
Shcherbakova, V.V., Biggin, A.J., Veselovskiy, R.V., Shatsillo, A.V., Hawkins, L.M.A., Shcherbakov, V.P., Zhidkov, G.V., 2017a. Was the Devonian geomagnetic field dipolar or multipolar? Palaeointensity studies of Devonian igneous rocks from the Minusa Basin (Siberia) and the Kola Peninsula dykes, Russia. Geophys. J. Int. 209. https://doi.org/10.1093/gji/ggx085.
Smirnov, A.V., Tarduno, J.A., Evans, D.A.D., 2011. Evolving core conditions ca. 2 billion years ago detected by paleosecular variation. Phys. Earth Planet. Inter. 187, 225–231. https://doi.org/10.1016/j.pepi.2011.05.003.
Smirnov, A.V., Tarduno, J.A., Kulakov, E.V., McEnroe, S.A., Bono, R.K., 2016. Palaeointensity, core thermal conductivity and the unknown age of the inner core. Geophys. J. Int. https://doi.org/10.1093/gji/ggw080.
Söderlund, U., Johansson, L., 2002. A simple way to extract baddeleyite (ZrO 2). Geochem., Geophys. Geosyst. https://doi.org/10.1029/2001GC000212.
Sprain, C.J., Swanson-Hysell, N.L., Fairchild, L.M., Gaastra, K., 2018. A field like today's? The strength of the geomagnetic field 1.1 billion years ago. Geophys. J. Int. 213, 1969–1983. https://doi.org/10.1093/gji/ggy074.
Stacey, J.S., Kramers, J.D., 1975. Approximation of terrestrial lead isotope evolution by a two-stage model. Earth Planet. Sci. Lett. https://doi.org/10.1016/0012-821X(75)90088-6.
Stepanova, A.V., Salnikova, E.B., Samsonov, A.V., Larionova, U.O., Arzamascev, A.A., Larionov, A.N., 2018. U-Pb geochronology of Early Precambrian mafic rocks in Kola-Murmansk province, Eastern Fennoscandia: dyke "bar-code" as the basis of paleocontinental reconstructions. In: Methods and Geological Results of Studying of Geochronometric Systems in Rocks and Minerals, pp. 340–343 (in Russian).
Svetov, S.A., Stepanova, A.V., Chazhengina, S.Y., Svetova, E.N., Rybnikova, Z.P., Mikhailova, A.I., Paramonov, A.S., Utitsyna, V.L., Ekhova, M.V., Kolodey, V.S., 2015. Precision geochemical (ICP-MS, LA-ICP-MS) analysis of rock and mineral composition: the method and accuracy estimation in the case study of early Precambrian mafic complexes. Prosideengs Karelian Res. Cent. RAS 173–192.
Tarduno, J.A., Cottrell, R.D., Smirnov, A.V., 2006. The paleomagnetism of single silicate crystals: recording geomagnetic field strength during mixed polarity intervals, superchrons, and inner core growth. Rev. Geophys. https://doi.org/10.1029/2005RG000189.
Tauxe, L., Banerjee, S.K., Butler, R.F., Voo, R. van der, 2009. Essentials of Paleomagnetism: Web Edition [WWW Document]. Scripps Inst. Oceanogr. URL < http://magician.ucsd.edu/essentials/WebBook.html > .
Tauxe, L., Shaar, R., Jonestrask, L., Swanson-Hysell, N.L., Minnett, R., Koppers, A.A.P., Constable, C.G., Jarboe, N., Gaastra, K., Fairchild, L., 2016. PmagPy: software package for paleomagnetic data analysis and a bridge to the Magnetics Information Consortium (MagIC) Database. Geochem., Geophys. Geosyst. https://doi.org/10.1002/2016GC006307.
Thellier, E., Thellier, O., 1959. Sur l'intensite du champ magnetique terrestre dans le passe historique et geologique. Ann. Geophys. 15, 285–376.
Timmerman, M.J., Daly, J.S., 1995. SmNd evidence for late Archaean crust formation in the Lapland-Kola Mobile Belt, Kola Peninsula, Russia and Norway. Precambrian Res. https://doi.org/10.1016/0301-9268(94)00045-S.
Torsvik, T.H., Meert, J.G., 1995. Early Proterozoic paleomagnetic data from the Pechenga zone (North-West Russia) and their bearing on early Proterozoic paleogeography. Geophys. J. Inter. 122, 520–536.
Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9. https://doi.org/10.1016/0040-1951(90)90116-P.
Veikkolainen, T.H., Biggin, A.J., Pesonen, L.J., Evans, D.A., Jarboe, N.A., 2017. Advancing Precambrian palaeomagnetism with the PALEOMAGIA and PINT(QPI) databases. Sci. Data. https://doi.org/10.1038/sdata.2017.68.
Veselovskiy, R.V., Arzamastsev, A.A., Demina, L.I., Travin, A.V., Botsyun, S.B., 2013. Paleomagnetism, geochronology, and magnetic mineralogy of Devonian dikes from the Kola alkaline province (NE Fennoscandian Shield). Izv. Phys. Solid Earth 49. https://doi.org/10.1134/S106935131303018X.
Veselovskiy, R.V., Arzamastsev, A.A., Tselmovich, V.A., Fetisova, A.M., Kulakova, E.P., 2017. Paleomagnetism of Precambrian dikes in the Kola part of northeastern Fennoscandia and its relation to the Svecofennian orogeny. Izv. Phys. Solid Earth 53. https://doi.org/10.1134/S1069351317050147.
Vetrin, V.R., 2014. Duration of the formation and sources of the granitoids of the Litsk-Araguba Complex, Kola Peninsula. Geochemistry Int. 52, 33–45.
Vodovozov, V.Y., Didenko, a.N., Gladkochub, D.P., Mazukabzov, a.M., Donskaya, T.V., 2007. Results of paleomagnetic study of Early Proterozoic rocks in the Baikal Range of the Siberian craton. Izv. Phys. Solid Earth 43, 855–867. https://doi.org/10.1134/S1069351307100084.
Widenfalk, L., Elming, S.Å., Enmark, T., 1985. A multidisciplinary investigation of the Notträsk Gabbro, northern Sweden. GFF. https://doi.org/10.1080/11035898509452623.
Wilson, R.L., 1961. The thermal demagnetization of natural magnetic moments in rocks. J. R. Astron. Soc. Geophys. https://doi.org/10.1111/j.1365-246X.1961.tb02928.x.
Yakubchuk, A., 2010. Restoring the supercontinent Columbia and tracing its fragments after its breakup: a new configuration and a Super-Horde hypothesis. J. Geodyn. 50, 166–175. https://doi.org/10.1016/j.jog.2010.03.001.
Yu, H.L., Zhang, L.F., Wei, C.J., Li, X.L., Guo, J.H., 2017. Age and P–T conditions of the Gridino-type eclogite in the Belomorian Province, Russia. J. Metamorph. Geol. https://doi.org/10.1111/jmg.12258.
Zhang, S., Li, Z.X., Evans, D.A.D., Wu, H., Li, H., Dong, J., 2012. Pre-Rodinia supercontinent Nuna shaping up: a global synthesis with new paleomagnetic results from North China. Earth Planet. Sci. Lett. https://doi.org/10.1016/j.epsl.2012.07.034.
Zhao, G., Cawood, P.A., Wilde, S.A., Sun, M., 2002. Review of global 2.1-1.8 Ga orogens: implications for a pre-Rodinia supercontinent. Earth-Sci. Rev. https://doi.org/10.1016/S0012-8252(02)00073-9.
Zhao, G., Sun, M., Wilde, S.A., Li, S., 2004. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth-Science Rev. 67, 91–123. https://doi.org/10.1016/j.earscirev.2004.02.003.